Quantitative risk assessment for data privacy: probability models, estimation uncertainty and governance thresholds

Ronald Rethabile Malamule Mahomane*

Department of Computing, Mathematical and Statistical Sciences, University of Namibia, Windhoek

School of Pathology, Faculty of Health Sciences, University of the Witwatersrand, Johannesburg

Basis Energy, Victoria, Australia

This version: September 2026

Abstract

Statistical agencies and central banks hold mature protection methods for confidential microdata but no agreed rule for deciding when a release may go out. This paper builds that rule in three separable parts. It first formulates re-identification as a probability conditional on a declared adversary, derives the prosecutor, journalist and marketer measures from one linkage experiment, and shows which assumptions the familiar reciprocal-frequency formulas hide. It then treats the risk figure as an estimate: we give the Monte Carlo estimator, the exact one-sided bound, the number of simulated attacks needed to certify a small tolerance, and the direction of the bias carried by plug-in frequency measures. Finally it treats the tolerance itself as a policy quantity with an owner, an expiry date and a declared harm class, and shows that the tolerance is not free: on a differentiable risk–utility frontier it is pinned by the price the institution places on harm relative to public value. Two results limit what a release programme can claim. Marginal risks bound the probability that some single release is attacked successfully, and the bounds are sharp; but an adversary who pools releases is not bounded by the marginals at all, and for any tolerance there exist releases that each satisfy it while the pooled attack succeeds with certainty. A simulation study on a synthetic register of 200,000 records and a 5 per cent sample reproduces each result, and a second study on a granular credit register measures five disclosure risks that a single lever cannot govern together, and the framework is then run on two real public microdata files, where 86 per cent of census records are unique on their key, the worst record in a 5-anonymous release is identified with probability 0.98, and four releases from one file admit a pooled attack at 1.86 times the bound their marginals imply. An adversarial evaluation then puts 7 protection policies that agencies operate, from the three-contributor rule to k-anonymity and noisy counts, against four attacks on a monthly credit release system, and reports them as attacks

*Corresponding author: ron@basisenergygroup.com. An earlier and shorter version was prepared for Session 5 of the Workshop on Data Security and Privacy, Banco de Portugal and Irving Fisher Committee, Lisbon, 14 and 15 September 2026. Every empirical table and quoted study result in this paper is produced by the packaged pipeline Mahomane [2026]; Section 10.10 describes what a run writes. The SENTRY-R example is a separate synthetic illustration.

are reported: success rate, precision on the attempts the adversary acts on, worst subgroup, and the auxiliary information the attack requires. Calibrated on one release, the least restrictive rule meeting a tolerance of 0.12 is no suppression at all; calibrated on the pooled programme it is a minimum cell count of 50, which cuts the file from 32,581 relationships to 20,389. Four releases that are each 3-anonymous, none containing a single unique record, admit a pooled attack that identifies the target with probability 0.46, which is 1.8 times the upper bound implied by the marginals. The practical recommendation is to publish, with every release decision, the risk estimate, its uncertainty, the tolerance, the adversary specification, the mechanism setting, the accountable owner and the expiry date. We additionally define a multidimensional trust position and a deep neural proposal model, SENTRY-R, that ranks mechanism settings across declared risk coordinates. Language models may propose scenarios for measured development attacks; independent simultaneous attack audits remain the decision gate. This extension is methodological and has a separate synthetic demonstration, not a measured improvement on the reported credit or census evaluations.

Keywords: statistical disclosure control; record linkage; re-identification risk; data governance; differential privacy; official statistics.

1 Introduction

Central banks, statistical offices and health registries release granular data under a statutory duty of confidentiality. The technical means of protection are well developed. Cell suppression, minimum cell counts, global recoding, record swapping, the cell-key method, additive noise, synthetic data and formal privacy mechanisms are all documented, implemented and taught [Willenborg and de Waal, 2001, Hundepool et al., 2012, Templ et al., 2015, Dwork and Roth, 2014]. The object that remains underdeveloped is the decision rule that turns a measured risk into an admissible or inadmissible release.

Practice usually settles this by committee. A disclosure control officer presents a table of risk measures, a panel weighs them against the public value of the release, and a decision follows. The decision is often sound. It is rarely reproducible, and it is hard to audit after the fact, because the quantity that did the work, namely how much residual risk the institution was willing to carry, was never written down. Improving the protection mechanism does not repair this. A mechanism with a formal guarantee can still be governed by an unstated tolerance, and an institution can adopt a carefully accounted ε without being able to say what probability of harm it accepted.

This paper separates three objects that are routinely run together.

1. A probability model for an adversary's attempted linkage, conditional on a declared attack scenario.
2. A statistical estimate of the resulting risk, with its uncertainty.
3. A policy decision that states how much residual risk is acceptable, for which harm class, on whose authority, and until when.

The first is a modelling problem, the second an inference problem, the third a governance problem. Confusing them produces two familiar failures. A committee that treats a point estimate as the risk releases at the boundary of its own tolerance about half the time. An institution that treats a protection parameter as a tolerance never states the probability it accepted.

1.1 Contribution

The contribution is mathematical as well as institutional.

First, we derive the prosecutor, journalist and marketer measures from a single linkage experiment (Section 3). The reciprocal-frequency formula $1/f_k$ is a theorem under four stated assumptions, not a definition of disclosure risk. Two of those assumptions, exchangeability of the records sharing a key and uniform selection among them, fail whenever the adversary holds informative side information, and then the posterior linkage probability exceeds $1/f_k$.

Second, we show that k-anonymity is exactly the decision rule $\sup_t R \leq 1/k$ under those four assumptions (Proposition 2), which gives the popular criterion a precise scope and a precise failure mode. In the simulation of Section 10, a 10-anonymous release contains a record whose

own re-identification probability is 0.74, with a one-sided 95 per cent lower bound of 0.64, and even the average risk over released targets, 0.1129, exceeds the nominal 0.100.

Third, we treat the risk figure as an estimate. We give the Monte Carlo estimator and its variance, the exact one-sided bound when no simulated attack succeeds, and the number of replicates needed to certify a tolerance (Proposition 3): certifying $\tau = 10^{-3}$ at the 5 per cent level needs 2,995 independent attacks, and $\tau = 10^{-5}$ needs 299,572. Observing no successful attack in a small experiment is not evidence of a small risk.

Fourth, we sign the bias of plug-in frequency measures (Proposition 4). Sample-frequency risk dominates population-frequency risk pathwise, so the two are not interchangeable summaries of the same release; in the simulation they differ by a factor of 7.3.

Fifth, we show that the tolerance is not a free parameter (Proposition 6). If the institution values utility at λ and harm at c_h, the admissible risk at the optimum is the point on the risk–utility frontier where the frontier slope equals the price ratio $\kappa = c_h/\lambda$. A fixed tolerance applied across releases is efficient only when the frontier slope at that tolerance is common to them (Corollary 7). The economics here is that of Abowd and Schmutte [2019]; what is new is its use as an operating rule for a release committee rather than as a social-choice argument.

Sixth, we settle what marginal risks can and cannot say about a release programme. The sharp bounds from the marginals are $\max_i R_i \le R_\cup \le \min(1, \sum_i R_i)$ (Proposition 9), and both ends are attainable. But the event that some single release is attacked successfully is not the event that the programme is attacked successfully. For an adversary who pools releases, no bound in terms of the marginals exists: for every $\nu > 0$ and every $m \ge 2$ there are m releases with marginal risk at most ν whose pooled attack succeeds with probability one (Proposition 10). The simulation shows it with ordinary keys, not a contrived construction.

Seventh, we relate scenario-conditional risk to formal privacy in a way that lets an institution hold both without comparing them on one scale (Proposition 14). Differential privacy bounds the movement of posterior odds; it does not name a linkage probability, and the linkage probability it permits depends on the prior.

Eighth, we extend certification to an adversary and an institution that may stop adaptively. A beta–binomial e-process gives a confidence sequence valid at every monitoring time (Proposition 13); a payoff model derives the posterior at which a rational adversary acts (Proposition 12); and a KL-neighbourhood gives an explicit upper bound under declared distribution shift. Each is measured in Section 8.1. An institution that watches its own simulation and stops when the fixed-sample bound first passes is wrong in 20 per cent of runs at a nominal 5 per cent; the confidence sequence costs 1.06 times the width and is not. An adversary that gains ten times what a false allegation costs it acts at a posterior of 0.136, not one half, and collects more than double the yield.

Ninth, we evaluate the framework as a system rather than as a formula. Seven protection policies that institutions operate are attacked on one monthly credit release programme, with the same adversary and the same auxiliary information, and reported the way attacks are reported: success rate, precision on the attempts the adversary acts on, the worst subgroup with a simultaneous bound, the auxiliary information the attack needs, and what happens when the adversary's own error model is wrong (Section 13). The evaluation prices Proposition 10.

At a tolerance of 0.12, per-release calibration selects no suppression and programme calibration selects a minimum cell count of 50, and the difference between them is the difference between 32,581 released relationships and 20,389.

Tenth, Section 14 defines a trust position over n declared adversary, group and vintage coordinates, gives an explicit neural prediction and independent audit protocol, and proves a simultaneous release guarantee for a fixed shortlist. The additional method is not used to generate the existing numerical results.

A simulation study on a synthetic register reproduces every result and supplies the numbers quoted above (Section 10). Section 11 runs the framework on two real public microdata files, and Section 12 applies it to five settings, starting with a granular credit register of the kind central banks collect, where identification, relationship disclosure, attribute inference, reconstruction from aggregates and differencing across monthly vintages are measured against one protection lever. Section 13 turns that case into an attack evaluation: the operational protection policies are put against the same four attacks on the same register, and per-release calibration is compared with programme calibration. Section 16 turns the framework into six falsification tests, each of which the simulation either passes or fails in a way that is informative. Section 17 states the record an institution should file with each release.

1.2 Relation to existing work

The measurement literature is long and we do not restate it. Disclosure risk as a probability conditional on an intruder scenario goes back to Duncan and Lambert [1989] and Elliot and Dale [1999]; the sample and population uniqueness measures and their estimation under a population-frequency model are due to Skinner and Elliot [2002], Reiter [2005] and Skinner and Shlomo [2008], with smoothing models developed by Rinott and Shlomo [2006] and earlier by Bethlehem et al. [1990]. Record linkage under error is Fellegi and Sunter [1969]. The risk–utility trade-off was formalised as a map by Duncan et al. [2001], with utility measures developed by Karr et al. [2006], Woo et al. [2009] and Snoke et al. [2018]. Formal privacy begins with Dinur and Nissim [2003] and Dwork [2006], Dwork et al. [2006], with the composition and accounting machinery in Dwork and Roth [2014], Bun and Steinke [2016], Mironov [2017] and Dong et al. [2022], and the statistical reading in Wasserman and Zhou [2010] and Duchi et al. [2018]. The economic reading of the trade-off as a social choice is Abowd and Schmutte [2019], with the census application in Abowd et al. [2022] and the counterargument in Hotz et al. [2022]. The governance literature, in particular the Five Safes [Desai et al., 2016] and the anonymisation decision-making framework [Elliot et al., 2020], is where tolerances and accountability are discussed, and it is largely disjoint from the measurement literature.

Three of this paper's components have close antecedents, and it is worth saying exactly what is taken from them. The result that a tolerance is pinned by the price the institution places on harm relative to public value is a restatement, in a per-release governance setting, of the social-choice argument of Abowd and Schmutte [2019]; the economics is theirs and the contribution here is only to carry it into a decision rule that a release committee can operate. The threshold at which an adversary acts follows from its payoffs rather than from the convention of one half, which is the premise of the game-theoretic analyses of re-identification risk [Wan et al., 2015];

what is added is the requirement that an evaluation report the precision and coverage curve over that threshold rather than one point on it. And the anytime-valid machinery of Section 8 is standard: the martingale bound is Ville [1939], the modern confidence sequences are Howard et al. [2021], and the framework is set out in Ramdas et al. [2023], with the adaptive-analysis problem in its statistical form in Dwork et al. [2015]. The contribution is the application, which is to certify a disclosure risk while the simulation is being monitored, and the measurement of what monitoring costs.

The two bodies of work are not joined. The measurement literature produces numbers without saying which decision they license; the governance literature produces accountability structures without saying which number triggers them. Two specific gaps motivate this paper. Neither literature treats the risk figure as an estimate with a stated error probability that enters the decision rule, though Brown et al. [2001] and the exact binomial interval of Clopper and Pearson [1934] give the required tools. And neither states what a programme of releases may claim, as opposed to a single release, once the adversary is allowed to combine them, though the reconstruction attacks of Dinur and Nissim [2003] and Garfinkel et al. [2019] and the de-anonymisation work of Narayanan and Shmatikov [2008] and Rocher et al. [2019] show that combination is the operative threat.

2 Population, sample, keys and mechanisms

Let the finite population be $P = \{1, \dots, N\}$. A confidential sample is indexed by the random set $I \subseteq P$ with $|I| = n$. Its inclusion indicators and first-order inclusion probabilities are

$$S_i = \mathbf{1}\{i \in I\}, \qquad \pi_i = \mathbb{P}(S_i = 1), \qquad i \in P. \tag{1}$$

Under simple random sampling without replacement, $\pi_i = n/N$ for every i; under Bernoulli sampling the S_i are independent with common π. A census is the limiting case $I = P$ and $\pi_i = 1$.

Let $X_i \in \mathcal{X}$ be the confidential record of unit i. An adversary does not necessarily observe all of X_i. It uses a declared key map

$$K_a : \mathcal{X} \longrightarrow \mathcal{K}_a, \tag{2}$$

where the subscript a records the assumed attack scenario. For $k \in \mathcal{K}_a$, define the population and sample key frequencies

$$F_k = \sum_{i \in P} \mathbf{1}\{K_a(X_i) = k\}, \qquad f_k = \sum_{i \in P} S_i \mathbf{1}\{K_a(X_i) = k\}. \tag{3}$$

Thus $0 \le f_k \le F_k$. The statements "sample unique" and "population unique" mean $f_k = 1$ and $F_k = 1$.

A protection mechanism is a Markov kernel $M_\theta(dy \mid x)$ indexed by $\theta \in \Theta$. The notation covers deterministic transformations as the degenerate case. Applied to the confidential sample

$X_I = (X_i : i \in I)$, it generates a release

$$Y_\theta \sim M_\theta(\cdot \mid X_I). \tag{4}$$

The randomness in (4) matters for everything that follows. Risk is a probability over four sources at once: the sampling design, the mechanism, the noisy or incomplete auxiliary information held by the adversary, and any randomisation in the adversary's own decision rule. An assessment that integrates over some of these and conditions on others must say which.

Remark 1. Nothing in this section requires θ to carry a formal guarantee. A minimum cell count of five, a coarser age classification and a Laplace mechanism with $\varepsilon = 1$ are all points in Θ. Section 9 explains why the last of these supports a claim the others do not, and why that claim is not a linkage probability.

3 A unified linkage experiment

Fix a target $t \in P$. The adversary observes auxiliary information Z_t and the release Y_θ, then applies a measurable linkage rule

$$\delta_a : \mathcal{Z} \times \mathcal{Y} \longrightarrow I \cup \{\emptyset\}. \tag{5}$$

The value $\delta_a(Z_t, Y_\theta) = j$ asserts that released record j belongs to t, and $\emptyset$ denotes abstention. Correct identification is the event

$$C_t(\theta, a) = \{S_t = 1,\ \delta_a(Z_t, Y_\theta) = t\}. \tag{6}$$

Definition 1 (Scenario-conditional target risk)**.** Given background information $\mathcal{B}_a$ declared by scenario a, the unconditional and file-membership-conditional risks are

$$R_a^{\mathrm{unc}}(t, \theta) = \mathbb{P}\{C_t(\theta, a) \mid \mathcal{B}_a\}, \qquad R_a^{\mathrm{in}}(t, \theta) = \mathbb{P}\{\delta_a(Z_t, Y_\theta) = t \mid S_t = 1, \mathcal{B}_a\}. \tag{7}$$

The distinction in (7) prevents a common error. If membership in the file is not known to the adversary, the unconditional risk carries an inclusion-probability factor,

$$R_a^{\mathrm{unc}}(t, \theta) = \pi_t \, R_a^{\mathrm{in}}(t, \theta), \tag{8}$$

provided a correct association is impossible when $S_t = 0$. Equation (8) is the multiplication rule $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B)$ and not an independence assumption. Reporting the two quantities under one label, which happens whenever a sampling fraction is quietly applied or quietly omitted, changes the reported risk by an order of magnitude in the simulation of Section 10.

3.1 Exact matching as a special case

Consider four assumptions.

Assumption 1. The released key is unperturbed: K_a applied to the release agrees with K_a applied to the confidential sample.

Assumption 2. The adversary observes the target's key without error.

Assumption 3. All records sharing the target's key are exchangeable in the adversary's information.

Assumption 4. The adversary selects uniformly among the compatible records.

Proposition 1 (Reciprocal frequency)**.** *Under Assumptions 1 to 4, conditional on* $S_t = 1$ *and* $K_a(X_t) = k$*,*

$$\mathbb{P}\{\delta_a = t \mid S_t = 1, K_a(X_t) = k, f_k\} = \frac{1}{f_k}. \tag{9}$$

Proof. See Appendix A. □

Proposition 1 is worth stating as a theorem because it is routinely used as a definition. Once it is a theorem, its assumptions can fail, and two of them fail often. Assumption 3 fails whenever the adversary holds a covariate correlated with the confidential variable, which is the normal case in financial and health data. Assumption 2 fails whenever the adversary's auxiliary source has reporting error, which is also the normal case.

3.2 Imperfect matching and posterior linkage probability

For perturbed keys or measurement error, let L_{tj} denote the likelihood of the observed auxiliary information and released record j under the hypothesis H_{tj} that j belongs to t, in the sense of Fellegi and Sunter [1969]. With prior match weights q_{tj} and a no-match hypothesis H_{t0}, Bayes' rule gives

$$p_{tj} := \mathbb{P}(H_{tj} \mid Z_t, Y_\theta, \mathcal{B}_a) = \frac{q_{tj} L_{tj}}{q_{t0} L_{t0} + \sum_{\ell \in I} q_{t\ell} L_{t\ell}}. \tag{10}$$

If the adversary chooses the maximum-posterior candidate, its conditional success probability is

$$R_a^{\text{in}}(t, \theta) = \mathbb{E}\Big[\max_{j \in I} p_{tj} \;\Big|\; S_t = 1, \mathcal{B}_a \Big]. \tag{11}$$

When the only compatible records have equal likelihood and the no-match prior is zero, (10) and (11) reduce to (9). Measurement error therefore cannot be declared to reduce risk. It changes the likelihood ratios, and it may either spread posterior mass across candidates or concentrate it on one. Section 10 exhibits a case where key reporting error lowers success from 0.4663 to 0.3812 while one noisy continuous covariate returns it to 0.4642, so that the reciprocal-frequency figure was right by cancellation rather than by argument.

3.3 The three conventional scenarios

Prosecutor. The prosecutor knows $S_t = 1$. Under exact exchangeable matching,

$$R_{\text{pros}}(t, \theta) = \frac{1}{f_{K_a(X_t)}}, \tag{12}$$

and a file-level conservative summary is the maximum target risk

$$R_{\max}(\theta) = \max_{i \in I} R_{\text{pros}}(i, \theta), \tag{13}$$

or the whole empirical risk distribution, which is more informative and no harder to publish. Calling (12) an upper bound over all conceivable scenarios requires Assumptions 2 to 4. An adversary with identifying side information breaks exchangeability and can achieve a posterior above $1/f_k$.

Journalist. The journalist does not know that the target is in the file. For a sample-unique released record with key k, the classical object is $\rho_k = \mathbb{P}(F_k = 1 \mid f_k = 1)$. Bayes' theorem gives

$$\rho_k = \frac{\mathbb{P}(f_k = 1 \mid F_k = 1)\mathbb{P}(F_k = 1)}{\sum_{v \geq 1} \mathbb{P}(f_k = 1 \mid F_k = v)\mathbb{P}(F_k = v)}. \tag{14}$$

Under Bernoulli sampling with common inclusion probability $\pi = n/N$,

$$\mathbb{P}(f_k = 1 \mid F_k = v) = \binom{v}{1} \pi (1 - \pi)^{v-1}, \tag{15}$$

and substitution into (14) yields

$$\rho_k = \frac{\pi \mathbb{P}(F_k = 1)}{\sum_{v \geq 1} v\pi(1 - \pi)^{v-1} \mathbb{P}(F_k = v)}. \tag{16}$$

Equation (16) shows why the sampling fraction matters, and it shows that π alone is not enough: the calculation needs a model or external evidence for the population frequency distribution [Skinner and Elliot, 2002, Reiter, 2005, Rinott and Shlomo, 2006]. Section 10 reports what that model costs. Using π itself as the journalist risk overstates the truth by a factor of about two at $\pi = 0.05$, while a fitted negative binomial frequency model, which is closed under binomial thinning and therefore identified from the sample, is accurate to within 8 to 25 per cent.

Marketer. The marketer attacks many targets and values the expected fraction correctly linked. If the attacked unit J is drawn uniformly from the population,

$$R^{\text{pop}}_{\text{mark}}(\theta) = \frac{1}{N} \sum_{i \in P} R^{\text{unc}}_a(i, \theta). \tag{17}$$

If instead one conditions on attacking the released records, and exact exchangeable matching holds, the sample mean is

$$R^{\text{sample}}_{\text{mark}}(\theta) = \frac{1}{n} \sum_{i \in I} \frac{1}{f_{K_a(X_i)}} = \frac{1}{n} \sum_{k : f_k > 0} 1 = \frac{D_I}{n}, \tag{18}$$

where D_I is the number of distinct released keys, because each occupied class contributes $f_k \cdot (1/f_k) = 1$. Formula (18) is not $n^{-1} \sum_i 1/F_{K(i)}$. The former models choice among released candidates; the latter presumes population frequencies are known and describes a different

experiment. The attack population and the conditioning event belong beside any marketer figure. In the simulation the sample-frequency measure is 7.3 times the population-frequency version, and the unconditional measure (17) is smaller than both, at 0.023 against 0.466 and 0.064.

3.4 k-anonymity is a threshold statement

Proposition 2 (k-anonymity as a decision rule)**.** *Under Assumptions 1 to 4, for $\tau \in (0, 1]$,*

$$\sup_{t \in I} R_{\text{pros}}(t, \theta) \leq \tau \quad \Longleftrightarrow \quad \min_{k: f_k > 0} f_k \geq \lceil 1/\tau \rceil, \tag{19}$$

so the release satisfies k-anonymity with $k = \lceil 1/\tau \rceil$ in the sense of Sweeney [2002]. If Assumption 3 or 4 fails, the implication from right to left fails: there are releases satisfying k-anonymity for which $\sup_t R_a^{\text{in}}(t, \theta) > 1/k$.

Proof. See Appendix A. The failure claim is constructive and is shown numerically in Section 10.4. □

Proposition 2 places k-anonymity where it belongs. It is not a rival to a probability model; it is the special case of the decision rule (29) below in which the risk model is exact exchangeable matching, the estimate carries no uncertainty, and the tolerance is the reciprocal of an integer. Each of those three simplifications is a place where an institution can be wrong, and the third is the only one that is a policy choice. The same reading applies to the syntactic strengthenings of k-anonymity, such as ℓ-diversity [Machanavajjhala et al., 2007] and t-closeness [Li et al., 2007]: they restrict the attribute distribution within a key class rather than the tolerance, and so they change the risk model, not the decision rule.

4 Estimation and statistical uncertainty

The risk in Definition 1 is unknown. It is a functional of the sampling design, the mechanism, the population frequency structure and the adversary's information, and in any realistic scenario it has no closed form. It must be estimated, and the estimate must carry its uncertainty into the decision.

4.1 Simulation estimator

Let $D^{(1)}, \ldots, D^{(B)}$ be attack simulations generated under the declared scenario, and set $C_t^{(b)} = 1$ if replicate b identifies t correctly. The Monte Carlo estimator is

$$\widehat{R}_a(t, \theta) = \frac{1}{B} \sum_{b=1}^{B} C_t^{(b)}, \qquad \mathbb{E}[\widehat{R}_a] = R_a, \qquad \text{Var}(\widehat{R}_a) = \frac{R_a(1 - R_a)}{B}, \tag{20}$$

the variance holding because the $C_t^{(b)}$ are independent Bernoulli variables under independent simulation. If replicates share auxiliary data, records or fitted parameters, independence fails and (20) understates the variance. The remedy is a clustered bootstrap over the shared unit,

a posterior interval, or a repeated-sample calculation that preserves the dependence. The clustered version resamples whole units of sharing, the auxiliary file or the fitted parameter draw, rather than individual attacks, and recomputes the risk within each resampled cluster [Efron and Tibshirani, 1993, van der Vaart, 1998]; the between-cluster spread then carries the dependence that (20) assumes away. The nested design of Section 10.9 is the same accounting written as an analysis of variance, and either device answers the same question: how much of the reported precision comes from the number of simulated attacks alone. Institutions that reuse one auxiliary file across all replicates are in this case and should not quote (20).

4.2 Small risks and the zero-success problem

For small risks a normal interval is unreliable [Brown et al., 2001]. An exact or score-based one-sided bound $\overline{R}_{1-\alpha}$ should be reported. With zero successes in B independent simulations, the one-sided Clopper–Pearson bound [Clopper and Pearson, 1934] solves $(1-U)^B = \alpha$, so

$$\overline{R}_{1-\alpha} = 1 - \alpha^{1/B} \approx -\frac{\log \alpha}{B}. \tag{21}$$

At 95 per cent confidence this is approximately $3/B$. Observing no successful attack does not establish zero risk, and the strength of the statement is set entirely by B.

Proposition 3 (Certification cost)**.** *Suppose B independent simulated attacks produce no success. The one-sided $(1-\alpha)$ bound satisfies $\overline{R}_{1-\alpha} \le \tau$ if and only if*

$$B \ge \frac{\log \alpha}{\log(1-\tau)} \approx \frac{-\log \alpha}{\tau} \quad \text{as } \tau \downarrow 0. \tag{22}$$

Proof. See Appendix A. □

The practical content of (22) is a budget. Certifying $\tau = 0.01$ at $\alpha = 0.05$ needs 299 independent attacks; $\tau = 10^{-3}$ needs 2,995; $\tau = 10^{-5}$ needs 299,572, and at $\alpha = 0.01$ the last figure becomes 460,515. Table 8 collects these. An institution that adopts a very small tolerance has thereby committed to a large simulation programme, or to a model-based argument in place of simulation. Choosing the tolerance and choosing the evidence standard are the same decision, and it is better made openly.

4.3 Plug-in frequency measures

Two reciprocal-frequency measures circulate, one built on sample frequencies and one on population frequencies. They are not two estimates of one quantity.

Proposition 4 (Ordering and plug-in bias)**.** *For every key k with $f_k > 0$ we have $1/f_k \ge 1/F_k$ pathwise, hence*

$$\mathbb{E}\Big[\frac{1}{f_{K(i)}}\Big] \ge \mathbb{E}\Big[\frac{1}{F_{K(i)}}\Big], \tag{23}$$

with equality only if the sample contains every population member of every occupied key class.

Moreover, for any random f_k that is positive with probability one and has finite mean,

$$\mathbb{E}\Big[\frac{1}{f_k}\Big] \geq \frac{1}{\mathbb{E}[f_k]}, \tag{24}$$

by Jensen's inequality, strictly unless f_k is degenerate. The reciprocal of an average frequency is therefore not an unbiased summary of average reciprocal-frequency risk, and the direction of the error is known.

Proof. See Appendix A. □

The gap is not a technicality. In the simulation of Section 10 the sample-frequency measure is 0.466, the population-frequency measure is 0.064, and the reciprocal of the mean sample frequency is 0.191. All three have been published as "the" re-identification risk of a microdata file. Only the first answers the question "if the adversary picks among the released candidates, how often does it win".

4.4 Model uncertainty

Simulation error is not the only uncertainty, and it is the only one the binomial expression describes. Let η index uncertain assumptions about population frequencies, measurement error and adversary capability, and define the robust risk

$$R_a^{\mathrm{rob}}(t, \theta) = \sup_{\eta \in H_a} R_a(t, \theta; \eta), \tag{25}$$

where the uncertainty set H_a is declared before evaluation. The order of operations decides whether this is a rule at all. Declaring H_a first and taking the supremum is a decision rule. Varying η after seeing the result and reporting the variant one prefers is sensitivity analysis presented as a decision rule, and it inherits none of the error control in Section 5.

It is useful to separate three layers of uncertainty that are otherwise hidden inside one error bar. Write $\widehat{\eta}$ for parameters estimated from an auxiliary source and $\mathcal{D}$ for the realised confidential sample. The law of total variance gives

$$\mathrm{Var}(\widehat{R}_a) = \mathbb{E}_{\mathcal{D},\widehat{\eta}}\{\mathrm{Var}(\widehat{R}_a \mid \mathcal{D}, \widehat{\eta})\} + \mathrm{Var}_{\mathcal{D},\widehat{\eta}}\{\mathbb{E}(\widehat{R}_a \mid \mathcal{D}, \widehat{\eta})\} = V_{\mathrm{MC}} + V_{\mathrm{input}}, \tag{26}$$

where the second term collects everything that varies with the sample and the fitted inputs rather than with the simulation. A nested design that redraws the sample estimates the part of it that the sampling design contributes. Separating that from the contribution of a refitted auxiliary model needs a second level of nesting, and the two parts do not in general add, because the risk is not separable in the two inputs. The binomial expression in (20) estimates only V_{MC}. Increasing B drives that term to zero and does nothing to the others.

For a smooth risk functional $R(\eta)$ with $\sqrt{m}(\widehat{\eta} - \eta_0) \Rightarrow N(0, \Sigma_\eta)$, the delta method supplies the first-order approximation [van der Vaart, 1998]

$$\mathrm{Var}_{\mathrm{aux}}\{R(\widehat{\eta})\} \approx \frac{1}{m} \nabla_\eta R(\eta_0)^\top \Sigma_\eta \nabla_\eta R(\eta_0). \tag{27}$$

Non-smooth operations such as a maximum over targets invalidate this calculation at ties, and a nested bootstrap or a simultaneous bound is then the safer object. Appendix B constructs both, and Section 10.9 measures the two terms of (26) on the simulated register.

5 Governance thresholds as decision rules

Definition 2 (Governance threshold)**.** A governance threshold is the tuple

$$G = (\tau, o, d, a, h, \alpha), \tag{28}$$

where $\tau \in (0, 1)$ is the risk tolerance, o is the named accountable owner, d is the expiry date, a is the adversary specification, h identifies the harm class to which the tolerance applies, and α is the accepted statistical error probability.

Each element earns its place by naming a failure it prevents. Without o the tolerance belongs to nobody and cannot be revised on authority. Without d an assessment made against a 2019 register governs a 2026 release. Without a the number has no referent, since risk is only defined given an adversary. Without h a tolerance calibrated for commercial nuisance is applied to a harm class where the consequence is physical. Without α the institution has not said how much evidence it requires.

Let $\overline{R}_{1-\alpha}(t, \theta)$ be an upper confidence or credible bound whose interpretation is recorded. A release is statistically admissible only if

$$\sup_{t \in T} \overline{R}_{1-\alpha}(t, \theta) \leq \tau, \tag{29}$$

the assessment date precedes d, and the release and adversary fall within the scope of a and h. The target set T must be explicit. A maximum over sample records, a maximum over a protected subgroup, and an average over a declared population are three different controls, and a release can pass one while failing another.

Proposition 5 (Why a point estimate is insufficient)**.** *If a rule releases whenever $\widehat{R}_a \leq \tau$, then at the boundary $R_a = \tau$ a consistent and asymptotically symmetric estimator releases with probability tending to* 1/2*. Replacing the point estimate by the one-sided rule* (29) *controls erroneous release at approximately α, subject to the interval model.*

Proof. See Appendix A. □

The two error rates are not symmetric in consequence, and an institution should state both. Rule (29) controls the probability of releasing when the true risk exceeds the tolerance at α. It pays for that control with erroneous suppression: a release whose true risk is comfortably below τ can still fail (29) when B is small, because $\overline{R}_{1-\alpha}$ is then far above $\widehat{R}$. In the simulation of Section 10, one setting on the frontier has a point estimate of 0.1058 from $B = 1{,}200$ replicates and a bound of 0.1216, so a tolerance of 0.12 rejects a release that its own point estimate accepts. Increasing B is the honest response, and Proposition 3 prices it.

5.1 The tolerance is not free

Definition 2 treats τ as a policy input. That is right in the sense that no statistical calculation can supply it, and wrong in the sense that once an institution has stated how it values public benefit against harm, the tolerance is implied. Let $U(\theta)$ be the utility of the release, defined in Section 6, and let the institution value utility at $\lambda > 0$ and expected harm at $c_h > 0$ per unit of identification probability, for the declared harm class h. Social value is

$$V(\theta) = \lambda U(\theta) - c_h R_a^{\mathrm{rob}}(\theta). \tag{30}$$

Proposition 6 (Implied tolerance)**.** *Suppose the risk–utility frontier of Section 6 is the graph of a differentiable, increasing, strictly concave function $u = \varphi(r)$ on an interval $[\underline{r}, \overline{r}]$. Then the maximiser of* (30) *over the frontier is interior and unique whenever $\varphi'(\underline{r}) > \kappa > \varphi'(\overline{r})$, where $\kappa = c_h/\lambda$, and it satisfies*

$$\varphi'(\tau^\star) = \kappa, \qquad \tau^\star := R_a^{\mathrm{rob}}(\theta^\star). \tag{31}$$

The implied tolerance $\tau^\star$ is decreasing in κ.

Proof. See Appendix A. □

Equation (30) is written with the robust risk. An institution that governs with the bound (29), as we recommend, should read (31) with $\overline{R}_{1-\alpha}$ in place of R_a^{rob}, and that is how the tolerance is computed in Section 10. The bound is a monotone transformation of the estimate at fixed B, so the ranking of settings and the frontier set are unchanged; κ is then a price per unit of bound rather than per unit of estimate, which is what an institution governing with (29) is in fact pricing.

Proposition 6 is the disclosure-control counterpart of the social choice framing in Abowd and Schmutte [2019]. Its practical use is not to compute τ, since κ is no easier to elicit than τ, but to expose the exchange rate a stated tolerance implies. An institution that adopts $\tau = 0.05$ on one release and $\tau = 0.20$ on another has priced harm differently in the two cases, and should be able to say why. In the simulation, moving κ from 0.25 to 10 moves $\tau^\star$ from 0.464 to 0.081 and changes the selected mechanism from an unsuppressed release to a 20-anonymous one that keeps 58 per cent of the records.

Corollary 7 (Uniform tolerances are usually inefficient)**.** *Let releases $i = 1, \dots, m$ have frontiers φ_i and a common price ratio κ. A common tolerance τ across the m releases is optimal only if $\varphi_i'(\tau) = \kappa$ for every i. Otherwise total value strictly increases by tightening the tolerance where the frontier is flat and loosening it where the frontier is steep.*

Proof. Immediate from (31) applied release by release. □

Two cautions belong with this result. The price ratio κ is not discovered in the data. Choosing it is a political and ethical act: it states what the institution believes an identification costs the person identified, relative to what the statistics are worth to everyone else, and no estimation procedure will supply that number. What the proposition does is force the choice into the open, because a tolerance already implies a κ whether or not anyone has said so. The

second caution follows. The accountable owner o of Definition 2 has to be senior enough to make that statement on the institution's behalf and be answerable for it, which in a central bank or a statistical office means the release committee or the executive member who chairs it, not the analyst who runs the simulation. An owner who cannot defend κ in public is not the right owner.

6 Utility and mechanism choice

Let $Q(P)$ be the estimand of interest and $\widehat{Q}(Y_\theta)$ its released-data estimator. A loss-based utility measure is

$$U(\theta) = -\mathbb{E}\big[\mathcal{L}\{\widehat{Q}(Y_\theta), Q(P)\}\big], \tag{32}$$

with expectation over sampling and mechanism randomness and $\mathcal{L}$ declared in advance. Squared error, absolute error, confidence-interval distortion and a weighted loss across authorised analyses are all admissible choices; the general and specific utility measures of Snoke et al. [2018] and the framework of Karr et al. [2006] fit here. Declaring $\mathcal{L}$ in advance matters for the same reason that declaring H_a in advance matters: a mechanism chosen after inspecting which loss it happens to minimise carries no guarantee.

The attainable set is

$$\mathcal{A} = \{(R_a^{\mathrm{rob}}(\theta), U(\theta)) : \theta \in \Theta\}. \tag{33}$$

A point $(r, u) \in \mathcal{A}$ is dominated if another attainable point (r', u') satisfies $r' \le r$ and $u' \ge u$ with at least one strict inequality. The risk–utility frontier is the nondominated subset

$$\mathcal{F} = \{(r, u) \in \mathcal{A} : \nexists (r', u') \in \mathcal{A} \text{ dominating } (r, u)\}, \tag{34}$$

which is the confidentiality map of Duncan et al. [2001] with a robust risk coordinate. Given the governed tolerance, mechanism selection is the constrained problem

$$\theta^\star \in \arg\max_{\theta \in \Theta} U(\theta) \quad \text{subject to} \quad \sup_{t \in T} \overline{R}_{1-\alpha}(t, \theta) \le \tau. \tag{35}$$

Proposition 8 (Existence)**.** *If Θ is compact, U is upper semicontinuous, $\theta \mapsto \sup_{t \in T} \overline{R}_{1-\alpha}(t, \theta)$ is lower semicontinuous and the feasible set is nonempty, then* (35) *has a solution.*

Proof. See Appendix A. □

The conditions in Proposition 8 fail for the discrete mechanism families used in practice, where Θ is a finite grid of suppression thresholds and recoding levels. Then the maximum over the grid exists trivially and the honest report is an approximate optimiser with the search resolution stated, not an implied global optimum. Section 10 reports a grid of 45 settings whose frontier contains 11 points.

One further caution belongs here. The frontier in (34) is itself estimated, and using the upper bound $\overline{R}_{1-\alpha}$ as the risk coordinate shifts every point rightward by an amount that depends on B. A frontier drawn with point estimates and a decision rule applied with bounds are not

the same object, and an institution that mixes them will choose mechanisms that its own rule rejects.

7 Release programmes and composition

Single-release assessment is not the operative problem. Institutions publish programmes: a quarterly series, a research file and a public extract from the same underlying register.

For releases $Y_1, \dots, Y_m$, let C_i be the event that release i, used alone, permits a correct identification under its declared attack, and let $R_i = \mathbb{P}(C_i)$. Write $C_\cup = \bigcup_{i=1}^m C_i$ and $R_\cup = \mathbb{P}(C_\cup)$.

Proposition 9 (Sharp bounds from marginal risks)**.** *Without assumptions on the dependence among the C_i,*

$$\max_i R_i \;\le\; R_\cup \;\le\; \min\Big(1, \sum_{i=1}^m R_i\Big), \tag{36}$$

and both bounds are attained for suitable joint distributions, so the marginals alone cannot identify a narrower universal interval.

Proof. See Appendix A. □

If the success events were independent one could write $R_\cup = 1 - \prod_i (1 - R_i)$, but repeated releases from the same underlying data are ordinarily dependent, and the expression must not be used without evidence for independence. In the simulation the independence value is 0.2313 against a Boole interval of $[0.0884, 0.2530]$, so it looks reassuringly close to the upper bound; that agreement is a coincidence of the design and carries no information.

7.1 Pooling defeats the marginals

Figure 1 shows the set operation behind a pooled attack. The small example uses two key classes of size three: each release alone leaves three candidates, while their intersection leaves one.

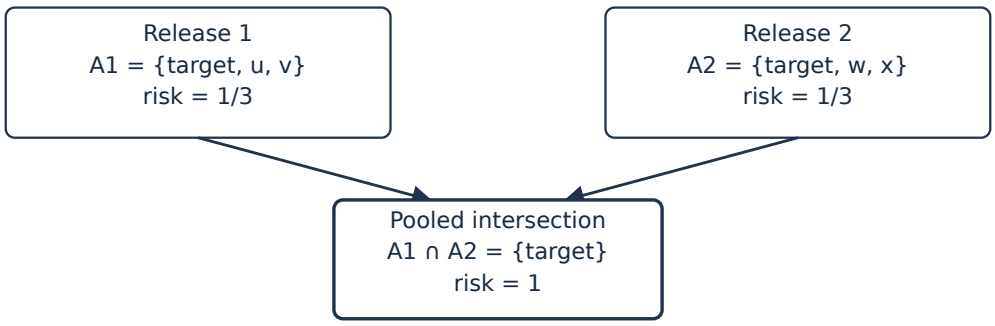


Figure 1: Pooling intersects candidate sets. This illustrative construction assumes uniform selection within each displayed set; it is distinct from the measured four-release simulation.

The deeper problem is that $C_\cup$ is the wrong event. An adversary who holds all m releases is not restricted to using them one at a time. It applies a joint linkage rule

$$\delta_a^{(m)} : \mathcal{Z} \times \mathcal{Y}_1 \times \cdots \times \mathcal{Y}_m \longrightarrow I \cup \{\emptyset\}, \tag{37}$$

and the programme-level risk is $R_{\text{pool}} = \mathbb{P}\{\delta_a^{(m)} = t\}$, which is not contained in the union $C_\cup$.

Proposition 10 (Marginal risks do not bound pooled risk)**.** *Let $m \geq 2$. For every $\nu > 0$ there exist a population, a sample and m releases such that $R_i \leq \nu$ for every i while $R_{\text{pool}} = 1$. Consequently no upper bound on R_{pool} can be written as a function of $(R_1, \ldots, R_m)$ alone. If each release publishes the candidate set implied by its key and the adversary selects uniformly from the intersection of those sets, then R_{pool} is nondecreasing in m and satisfies $R_{\text{pool}} \geq \max_i R_i$.*

Proof. See Appendix A. The construction takes $n = k^m$ units labelled by vectors in $\{1, \ldots, k\}^m$ and lets release i key on all coordinates except the ith, so that every release is k-anonymous while the intersection of the m candidate sets is a singleton. □

Proposition 10 does not depend on the construction. Section 10.6 builds four releases from ordinary demographic keys, each carrying a minimum cell count of three and containing no unique record at all, and the pooled attack identifies the target with probability 0.4581 against a Boole upper bound of 0.2530. Two releases are already enough: the pooled risk at $m = 2$ is 0.4114, which is 1.6 times the bound the four marginals imply.

Neither the phenomenon nor its direction is new. Ganta et al. [2008] showed that independent releases of overlapping populations, each protected by a syntactic rule, can be combined into a successful attack, and the failure of syntactic criteria under composition is the standard argument for a formal accountant [Dwork, 2006, Dwork and Roth, 2014]. What Proposition 10 adds is a statement in the language institutions actually use to certify releases. The composition-attack literature shows that combination defeats a syntactic rule; the proposition says that no function of the published marginal risks can bound the combined risk, so the quantity a release programme is certified against is not the quantity that governs it. Section 13 then prices that gap on a release system with an operational lever, which is the part of the argument the earlier work does not supply.

Three consequences follow for practice. Marginal risk figures may be published, but they license no programme-level claim. A programme tolerance must be assessed by direct joint simulation under (37), with shared targets and shared auxiliary information; separate simulations of each release cannot recover cross-release dependence by construction, because they never form the intersection. And a formal accountant, discussed in Section 9, is the one device that does compose without a dependence assumption.

7.2 Allocating a programme tolerance

Suppose an institution accepts the conservative route and governs the union event, so that the programme constraint is $\sum_i R_i \leq \tau_{\text{prog}}$ by (36). Each release has value weight $w_i > 0$ and frontier φ shared across releases for simplicity.

Proposition 11 (Risk budget)**.** *The problem* $\max \sum_i w_i \varphi(r_i)$ *subject to* $\sum_i r_i \le \tau_{\text{prog}}$ *and* $r_i \ge 0$*, with* φ *increasing, differentiable and strictly concave, and with* $\varphi'(0^+) = \infty$*, so that no release is allocated zero risk, has a unique solution characterised by*

$$w_i \varphi'(r_i^\star) = \mu \quad \text{for all } i, \qquad \sum_i r_i^\star = \tau_{\text{prog}}, \tag{38}$$

for a multiplier $\mu > 0$*. If* $\varphi(r) = \log r$ *then* $r_i^\star = \tau_{\text{prog}} w_i / \sum_j w_j$*, so the budget is allocated in proportion to value.*

Proof. See Appendix A. □

The equal split $r_i = \tau_{\text{prog}}/m$, which is what an institution does by default when it applies one tolerance to every release, satisfies (38) only if all w_i are equal. The proportional rule is equally particular: it holds for the logarithmic value curve and for no other. For the power family $\varphi(r) = a + br^\gamma$ with $b < 0$ and $\gamma < 0$, which is the family that fits the simulated frontier, condition (38) gives

$$r_i^\star \;\propto\; w_i^{1/(1-\gamma)}, \tag{39}$$

so the allocation is flatter than value whenever $\gamma < 0$, and it approaches an equal split as $\gamma \to -\infty$. In the simulation the fitted $\gamma = -0.23$ gives an exponent of 0.81, and the resulting allocation cuts the total loss by 4.5 per cent relative to the equal split.

We stress the scope. Proposition 11 governs the union event. It does not govern the pooled attack, and by Proposition 10 nothing stated in terms of the marginals can. A risk budget is a device for allocating attention across releases, not a substitute for joint simulation.

8 Adaptive attacks and anytime-valid certification

The preceding analysis fixes the number of releases and attacks. An operational adversary does neither: it observes a new vintage, decides whether to acquire more auxiliary information, and stops when the posterior is sufficiently favourable. An institution also monitors its simulation and may be tempted to stop when the bound first passes. Fixed-sample intervals do not generally survive either form of optional stopping.

Let $\mathcal{F}_s = \sigma(Y_1, \dots, Y_s, Z^{(s)})$ be the information available after release s, including auxiliary information acquired adaptively, and let H_t be the hypothesis that candidate record j belongs to target t. The posterior odds process is

$$O_s(H_t) = O_0(H_t) \prod_{v=1}^{s} \frac{p(Y_v, Z^{(v)} \mid H_t, \mathcal{F}_{v-1})}{p(Y_v, Z^{(v)} \mid H_t^c, \mathcal{F}_{v-1})}. \tag{40}$$

The conditional densities in (40) prevent the common error of multiplying marginal likelihoods from dependent vintages. Revisions, common records and mechanism randomness shared across releases must remain in the conditioning history.

An adversary need not use the arbitrary posterior threshold one half. Suppose a correct allegation produces benefit $h > 0$, a false allegation costs $c_f \ge 0$, and making the allegation costs $c_a \ge 0$. At posterior match probability p, the expected payoff is $ph - (1-p)c_f - c_a$.

Proposition 12 (Economically rational attack threshold)**.** *An expected-payoff-maximising adversary acts if and only if*

$$p \geq p^\star := \min\left\{1, \frac{c_f + c_a}{h + c_f}\right\}. \tag{41}$$

The threshold equals one half only in the special case $h = c_f$ and $c_a = 0$.

The attack evaluation must therefore report the precision–coverage curve over $p^\star$, or state the payoff assumptions that select one point. A low average success rate can coexist with a high-payoff attack that makes few, accurate allegations.

For institutional monitoring, let $X_b \in \{0, 1\}$ be success in simulated attack b, $S_B = \sum_{b \leq B} X_b$, and consider a candidate risk $r \in (0, 1)$. Mixing the Bernoulli likelihood ratio against a $\mathrm{Beta}(a, b)$ alternative gives the e-process

$$E_B(r) = \frac{\mathrm{B}(a + S_B, b + B - S_B)}{\mathrm{B}(a, b) r^{S_B} (1 - r)^{B - S_B}}. \tag{42}$$

Proposition 13 (Anytime-valid risk sequence)**.** *If the simulated attacks are conditionally Bernoulli with risk r_0, then $\{E_B(r_0), \mathcal{F}_B\}_{B \geq 0}$ is a nonnegative martingale with initial value one. Consequently*

$$\mathbb{P}_{r_0}\left\{\sup_{B \geq 1} E_B(r_0) \geq \alpha^{-1}\right\} \leq \alpha. \tag{43}$$

The confidence sequence

$$\mathcal{C}_B = \{r \in (0, 1) : E_B(r) < \alpha^{-1}\} \tag{44}$$

therefore contains r_0 simultaneously for every monitoring time with probability at least $1 - \alpha$ [Ville, 1939, Howard et al., 2021].

For q targets, mechanisms or subgroups selected after inspection, replace α by α/q in (44); the union bound then gives simultaneous anytime coverage without independence. This is conservative but valid. A hierarchical or max-e-process can recover power only if its dependence model is declared before the results are observed.

Finally, calibration on one attack distribution P does not transport without a condition on the deployment distribution Q. For the success indicator $0 \leq C \leq 1$,

$$R_Q - R_P = \mathbb{E}_Q C - \mathbb{E}_P C \leq \|Q - P\|_{\mathrm{TV}} \leq \sqrt{\frac{1}{2} D_{\mathrm{KL}}(Q \| P)}, \tag{45}$$

where the second inequality is Pinsker's inequality. Thus a declared KL-neighbourhood $D_{\mathrm{KL}}(Q\|P) \leq \rho$ gives the transport bound $R_Q \leq \min\{1, R_P + \sqrt{\rho/2}\}$. This is not a claim that KL balls describe every future adversary. It is a computable sensitivity statement, and the pipeline implements (42)–(45), so optional stopping and distribution shift are measured here rather than listed as verbal limitations.

8.1 What the pipeline measures

The three devices above are implemented in the packaged pipeline, and this subsection reports what they give on the release system of Section 13. The reference attack throughout is the

Table 1: The cost of watching the run. The fixed-sample column is the number of successless attacks that certifies the tolerance at one prearranged sample size. The anytime column is the number that certifies it while the institution monitors and may stop whenever it likes, using the confidence sequence of Proposition 13. The last column is the ratio, which is the price of the right to stop early.

Tolerance τ	α	Fixed sample	Anytime	Ratio
0.01	0.05	299	984	3.3
0.001	0.05	2,995	12,417	4.1

Table 2: Where a paid adversary acts. The payoff of a correct allegation, the cost of a false one and the cost of making any allegation fix the posterior threshold of Proposition 12. Coverage is the share of targets the adversary acts on at that threshold, precision the share of those allegations that are true, and yield the product, which is what the adversary collects. The attack is the pooled one against k anonymity 5. One half, the threshold usually assumed, is the middle case of equal stakes and free allegations.

Benefit h	False cost c_f	Action cost c_a	$p^\star$	Coverage	Precision	Yield
1	1	0	0.500	0.114	0.739	0.084
10	1	0.5	0.136	0.355	0.463	0.165
1	5	0.5	0.917	0.034	0.978	0.033

pooled one against the 5-anonymous release, whose success rate is 0.224.

Monitoring is not free, and pretending otherwise is worse. Take an institution that simulates 4,000 attacks, looks at its bound 20 times along the way, and stops the moment the bound falls below the truth. Over 500 such runs the fixed-sample bound at the 5 per cent level does that in 20 per cent of them. The confidence sequence (44) does it in 0 per cent. The stated level of a fixed-sample interval is a statement about one prearranged sample size, and an institution that watches its own simulation is not making that statement.

The price of the right to stop is a wider interval, and it is small. On the same counts the fixed-sample upper bound is 0.235 and the anytime bound is 0.249, a ratio of 1.06 over the runs of the study.

The certification cost of Proposition 3 rises in the same way. Table 1 gives both counts. Certifying a tolerance of 10^{-3} at the 5 per cent level needs 2,995 successless attacks when the sample size is fixed in advance, and 12,417 when the run is monitored throughout. An institution that wants to stop early should budget for the second number, not discover it later.

The adversary's threshold is a payoff, not a convention. Table 2 reads the same attack at the thresholds three payoff structures imply. An adversary that gains and loses equally and pays nothing to allege acts at one half, and collects a yield of 0.084 successful allegations per target. An adversary that gains ten times what a false allegation costs it acts at 0.136 instead: it acts on 0.355 of targets, is right 0.463 of the time, and its yield is 0.165, more than double the first. An adversary facing reputational or legal cost for a false allegation acts at 0.917, on 0.034 of targets, and is right 0.978 of the time.

The three rows are the same attack on the same release. An evaluation that reports one threshold reports one row of a curve the adversary chooses from, and an institution that assumes one half is assuming a payoff structure it has not stated. The precision–coverage curve belongs in the release record with the risk estimate.

Shift has a price the institution can name. The transport bound (45) turns a declared neighbourhood into a number. At a divergence of 0.05 the risk of the reference attack is bounded by 0.382, against 0.274 at the smallest neighbourhood in the table. Read the other way, and this is the more useful direction, the bound says how much shift a release can absorb before it breaks its tolerance. The reference release has none to spare: its pooled risk already exceeds the tolerance of 0.12, so any neighbourhood at all leaves it inadmissible. The one release in the comparison set that does meet the tolerance pooled, noisy counts, $\varepsilon = 1$ at a pooled risk of 0.000, can absorb a divergence of 0.029 and no more. That is a small neighbourhood, and saying so is an honest statement of how much the certificate depends on the adversary specification, which the usual sensitivity paragraph leaves unquantified.

9 Formal privacy is a different regime

A randomised mechanism M is $(\varepsilon, \delta_{\mathrm{dp}})$-differentially private if, for all neighbouring datasets $x \sim x'$ and all measurable output sets E,

$$\mathbb{P}\{M(x) \in E\} \le e^{\varepsilon}\, \mathbb{P}\{M(x') \in E\} + \delta_{\mathrm{dp}}. \tag{46}$$

For mechanisms M_i with parameters $(\varepsilon_i, \delta_{\mathrm{dp},i})$, basic adaptive composition gives $\varepsilon_{\mathrm{tot}} = \sum_i \varepsilon_i$ and $\delta_{\mathrm{dp,tot}} = \sum_i \delta_{\mathrm{dp},i}$, and tighter accountants are available [Bun and Steinke, 2016, Mironov, 2017, Dong et al., 2022]. Even the basic form shows the structural difference from Section 7: the composition holds without any assumption about the adversary or about dependence between releases.

The parameter ε bounds the change in output distributions for neighbouring databases. It is not the probability that an identified person is correctly linked, and the two must not be compared on a common numerical scale. The relation that does hold is Bayesian and depends on the prior.

Proposition 14 (What ε permits)**.** *Let M satisfy* (46) *with $\delta_{\mathrm{dp}} = 0$, and let H be a hypothesis about a single individual such that the databases x and x' under H and its negation differ only in that individual's record, the rest of the database having the same distribution under both. For a Bayesian adversary with prior $\mathbb{P}(H) = p$, the posterior after observing that the release falls in a measurable set E of positive probability satisfies*

$$\mathbb{P}(H \mid M \in E) \;\le\; \frac{e^{\varepsilon} p}{e^{\varepsilon} p + (1 - p)}. \tag{47}$$

Proof. See Appendix A; the argument is the Bayesian reading of differential privacy in Kasiviswanathan and Smith [2014]. □

Inequality (47) is the honest translation, and it shows why no single number does both jobs. With a prior of 0.1, which is what a 10-anonymous key class supplies under exact exchangeable matching, $\varepsilon = 0.5$ permits a posterior of 0.155, $\varepsilon = 1$ permits 0.232 and $\varepsilon = 2$ permits 0.451. With a prior of 1/3 the same $\varepsilon = 1$ permits 0.576. An institution that adopts $\varepsilon = 1$ has not chosen a tolerance of 1; it has chosen a multiplier on whatever prior the adversary brings, and the tolerance it has implicitly accepted depends on the key structure of the file.

The two regimes therefore answer different questions and both belong in the record. Scenario-conditional risk answers what a named adversary would achieve against this release, which is the question a governance committee, a data protection authority and an affected person all ask. Formal privacy answers what any adversary could learn from the mechanism across a programme, which is the question that scenario analysis cannot answer because Proposition 10 denies it a composition theorem. Kifer and Machanavajjhala [2011] shows that neither can be made assumption-free. The practical arrangement is to run a formal accountant for the programme budget and a scenario assessment for the release decision, and to report them separately with their assumptions attached.

The distinction is worth stating in one line, because it decides which tool an institution should reach for. The framework of this paper is adversary-specific: it answers what happens if someone with these resources attacks this release, and the answer is a probability that can be compared with a tolerance and audited against a stated attacker. Differential privacy is adversary-agnostic: it answers how much any observer's beliefs can move because one person's record was included, whoever that observer is and whatever it knows, and the answer is not a linkage probability. The first is what a release committee needs, because a committee must justify a decision about a particular publication to a particular public. The second is what a programme needs, because it is the only device that composes without an assumption about how releases relate to one another. An institution that wants both should hold both, and should not attempt to convert one number into the other.

10 Simulation study

The results above are statements about any release. This section applies them to a synthetic register where the population is known, so that every estimate can be compared with a truth. All quantities come from one program with fixed seeds; Appendix C gives the details.

10.1 Design

The population has $N = 200{,}000$ units. Each unit carries a district (40 categories with a skewed size distribution), sex, age in single years from 18 to 87, and an occupation group (12 categories, also skewed). The adversary's key is the cross-classification of district, sex, five-year age band and occupation group, giving 13,440 possible cells of which 12,596 are occupied. 1,305 cells are population unique, so 0.65 per cent of the population would be identifiable from the key alone in a census.

The confidential variable is

$$y_i = 8 + \psi_{d(i)} + \omega_{o(i)} + 0.018\,(\mathrm{age}_i - 50) + 0.10\,\mathrm{sex}_i + \epsilon_i, \qquad \epsilon_i \sim N(0, 0.6^2), \tag{48}$$

with district and occupation effects drawn once from $N(0, 0.45^2)$ and $N(0, 0.35^2)$. A Bernoulli sample with $\pi = 0.05$ gives $n = 10{,}035$ records containing 4,679 distinct keys, of which 2,729 are unique, so 27.2 per cent of the released records are sample unique.

The release supports three declared analyses: the mean of y by district, by five-year age band, and by occupation. Utility (32) is the negative of the average of the three root mean squared errors, each computed at the finest classification its analysis needs and normalised by the standard deviation of y, plus a penalty of 0.5 times the share of target categories the release leaves unreportable. Coarsening the key therefore costs accuracy on the analysis it coarsens, and suppression costs coverage, so both levers have a price. The two interact: once a minimum cell count binds, coarsening also buys back records, so the levers are substitutes as well as costs, which is why the frontier has to be traced rather than reasoned out.

The declared adversary observes district and sex without error, the released age band with probability 0.15 of a one-band error, the occupation group with probability 0.10 of a wrong group, and a noisy reading $z_t = y_t + N(0, 0.75^2)$ of the confidential variable. It links by maximum posterior under the same error model that generates the error, which is (10) with a flat prior over the records in the target's district and sex. The uncertainty set H_a in (25) contains two adversaries, the declared one and one whose covariate is twice as precise, each modelling its own information correctly. The robust risk is the larger, and across all 45 settings of the grid it is attained by the more capable adversary, whose risk runs from 1 to 76 per cent above the declared one, with a median gap of 40 per cent.

Two target sets are reported, because as Section 5 insists they are different controls. The first is a target drawn uniformly from the released records, whose risk is an average; this is the coordinate used for the frontier and the mechanism grid. The second is the set of records sitting in the smallest released cells, whose risk is estimated target by target and summarised by its maximum; this is the supremum control of (29) and it is the stricter of the two.

The mechanism family is a grid of 45 settings: age band width in $\{5, 10, 20\}$ years, occupation groups in $\{12, 6, 3\}$, and minimum cell count $k_{\min} \in \{1, 3, 5, 10, 20\}$ with records in smaller cells suppressed. A district record swapping lever was built and then removed, for a reason that belongs in the paper rather than a footnote. Swapping reduced the simulated risk, but only against an adversary that takes the released district at face value, and the declared scenario gives no reason to assume that. Its apparent protection was a property of the assumed adversary rather than of the release, which is the failure that Section 4 warns about, and the honest response was to drop the lever rather than to report a risk figure that depends on it.

10.2 The conventional measures disagree by orders of magnitude

Table 3 reports the standard summaries for the unprotected release. The identity (18) holds exactly: the mean sample-frequency risk and D_I/n are the same number, 0.4663. The population-frequency version is 0.0636, and the unconditional measure (17), which carries the inclusion prob-

Table 3: Conventional risk summaries for the unprotected release ($N = 200{,}000$, $\pi = 0.05$, $n = 10{,}035$).

Measure	Experiment described	Value
Prosecutor, mean $1/f_k$	uniform choice among released candidates	0.4663
Prosecutor, max $1/f_k$	worst released record	1.0000
Share of records with $f_k = 1$	sample uniqueness	0.2719
Marketer, D_I/n	identity (18), same experiment	0.4663
Plug-in $n^{-1}\sum_i 1/F_{K(i)}$	population frequencies assumed known	0.0636
Marketer, unconditional (17)	membership unknown, π applied	0.0233
π times the plug-in measure	neither experiment; reported anyway	0.0032
Journalist ρ, truth	$\mathbb{P}(F_k = 1 \mid f_k = 1)$, computed	0.0227
Journalist ρ, model-based	fitted negative binomial for F_k	0.0274
Journalist, naive π	sampling fraction used as the risk	0.0500

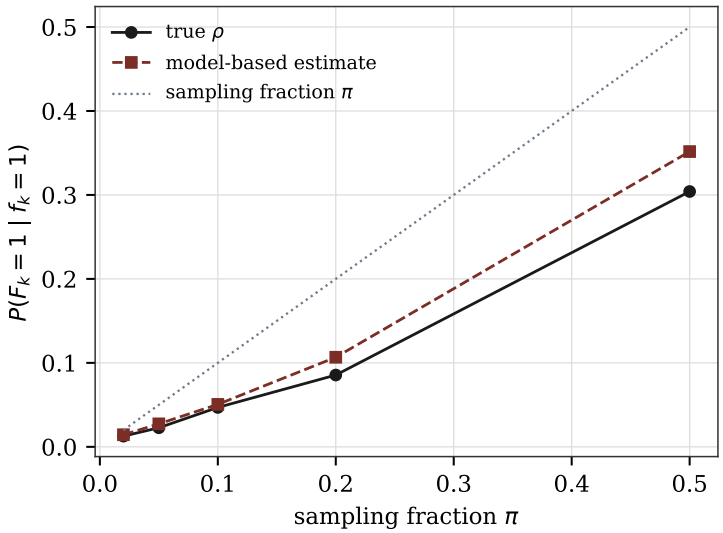


Figure 2: Journalist risk $\mathbb{P}(F_k = 1 \mid f_k = 1)$ against the sampling fraction. The model-based estimate uses a negative binomial population frequency law fitted to the sample cell counts. The sampling fraction itself is not an approximation to the risk.

ability of (8), is 0.0233. The sample-frequency measure is 7.3 times the population-frequency version and 20 times the unconditional one, and all three have been published under the same label.

The journalist row deserves attention because it is the one measure that needs a model for an unobservable. Since binomial thinning of a negative binomial law is negative binomial with the same dispersion, the population frequency law is identified from the sample cell counts, and the fit recovers a mean population cell size of 14.83 against a truth of 14.88, within 1 per cent. Figure 2 compares the resulting ρ with the truth across sampling fractions. The model-based estimate is conservative at every sampling fraction and its relative error lies between 8 and 25 per cent. Using π itself as the risk, which is a common shortcut, overstates the truth at every sampling fraction, by a factor of 2.20 at $\pi = 0.05$ and 1.64 at $\pi = 0.5$.

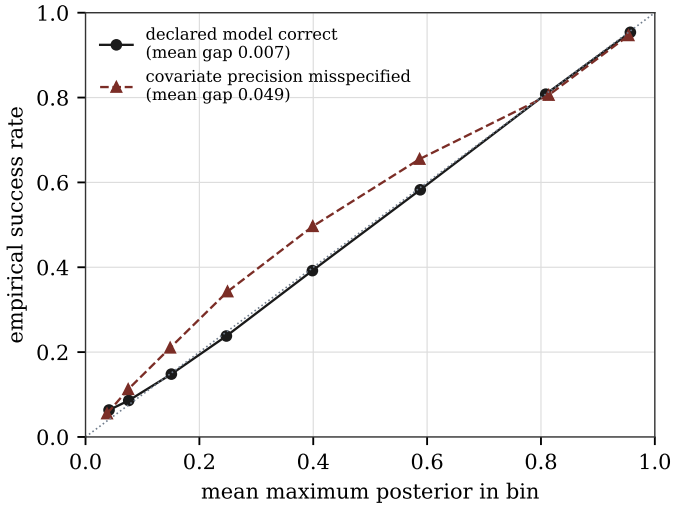


Figure 3: Calibration of the adversary's maximum posterior against the empirical success rate, 20,000 replicates per curve. The two models produce risk estimates that differ by 0.007, while their calibration gaps differ by a factor of seven.

10.3 Measurement error does not simply reduce risk

Under exact exchangeable matching the predicted success rate for a uniformly drawn released target is 0.4663. Simulating the declared adversary with key reporting error but no covariate gives 0.3812 over 5,000 replicates, which is the reduction practitioners expect. Adding the single noisy covariate returns the success rate to 0.4642 over 20,000 replicates, with a one-sided 95 per cent bound of 0.4701. The reciprocal-frequency figure is therefore accurate for this release, but by cancellation between an error that spreads posterior mass and a covariate that concentrates it. An assessment that modelled only the reporting error would have understated the risk by 18 per cent.

Figure 3 tests the posterior itself, which is the first falsification requirement of Section 16. When the adversary's declared model matches the process that generates its information, the mean gap between the binned maximum posterior and the empirical success rate is 0.007. When the adversary assumes a covariate 2.5 times noisier than the truth, the mean gap rises to 0.049 and the miscalibration is one-sided: the posterior understates the success rate in the lower bins. The overall success rate barely moves, from 0.4642 to 0.4571. This is the practical lesson of the figure. A single risk number cannot reveal that the linkage model is wrong, and a calibration check on held-out matches can.

10.4 k-anonymity does not deliver $1/k$

Table 4 applies minimum cell counts and compares the nominal bound $1/k$ with two simulated quantities: the average risk over a uniformly drawn released target, and the risk of the worst target among those in the smallest released cells, each of the latter estimated from 250 replicates.

The average risk stays below $1/k$ at $k = 3$ and $k = 5$ and exceeds it at $k = 10$ and $k = 20$. The worst target's risk exceeds $1/k$ at every k, and so does its one-sided 95 per cent lower confidence bound, which is 0.82, 0.78, 0.64 and 0.68 as k runs from 3 to 20. Those bounds carry

Table 4: Minimum cell counts against the simulated risk of the declared adversary. The average uses 12,000 uniformly drawn targets with $\overline{R}_{0.95}$ its one-sided Clopper–Pearson bound. The worst target is taken over the listed number of targets in minimum-size cells, each estimated from 250 replicates, and is reported with a one-sided 95 per cent lower bound carrying a Bonferroni correction for the number of targets examined. The last column is the share of those targets whose individual lower bound exceeds $1/k$.

$k_{\min}$	Kept	Share	$1/k$	Average	$\overline{R}_{0.95}$	Targets	Worst (bound)	Share above $1/k$
3	5,426	0.541	0.3333	0.2573	0.2640	250	0.900 (0.817)	0.748
5	3,361	0.335	0.2000	0.1852	0.1911	250	0.868 (0.778)	0.492
10	1,555	0.155	0.1000	0.1129	0.1178	230	0.744 (0.638)	0.326
20	484	0.048	0.0500	0.0787	0.0828	60	0.768 (0.675)	0.267

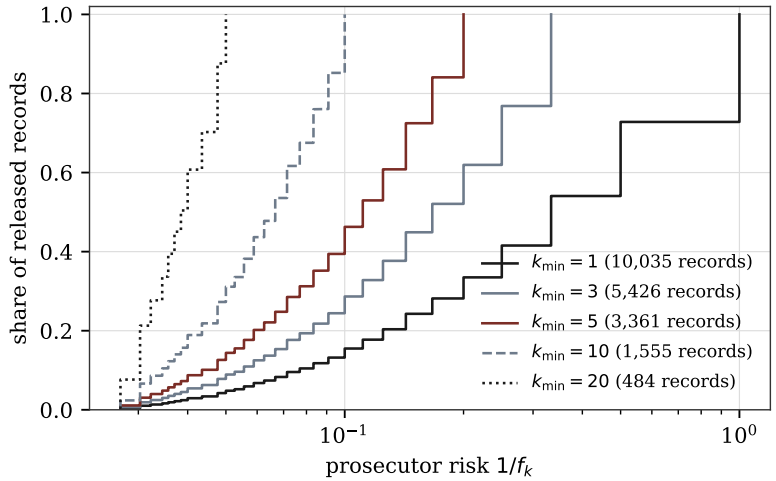


Figure 4: Distribution of prosecutor risk $1/f_k$ across released records under minimum cell counts. Governing the maximum alone forces most of the file out of the release.

a Bonferroni correction for the number of targets examined, since the worst target is selected on its own estimate. Between 27 and 75 per cent of the targets in minimum-size cells have an individual lower bound above $1/k$. Both comparisons refute the reading of k-anonymity as a guarantee that no released person faces a risk above $1/k$, and the second refutes it in the form that matters for rule (29), which is stated as a supremum. The mechanism is the one Proposition 2 names: the covariate separates members of a key class, and raising k shrinks the share of the adversary's information that the criterion controls while leaving the covariate untouched.

Figure 4 shows what the suppression does to the whole risk distribution rather than to its maximum, and the price is high. Reaching $k = 10$ retains 15.5 per cent of the released records and reaching $k = 20$ retains 4.8 per cent, so a release governed by its maximum prosecutor risk alone is pushed towards discarding most of its own content.

10.5 Frontier, tolerance and the price of harm

Figure 5 plots all 45 mechanism settings in the plane of the one-sided risk bound and the utility of Section 6. Eleven settings are nondominated, and the frontier spans risk from 0.081 to 0.564.

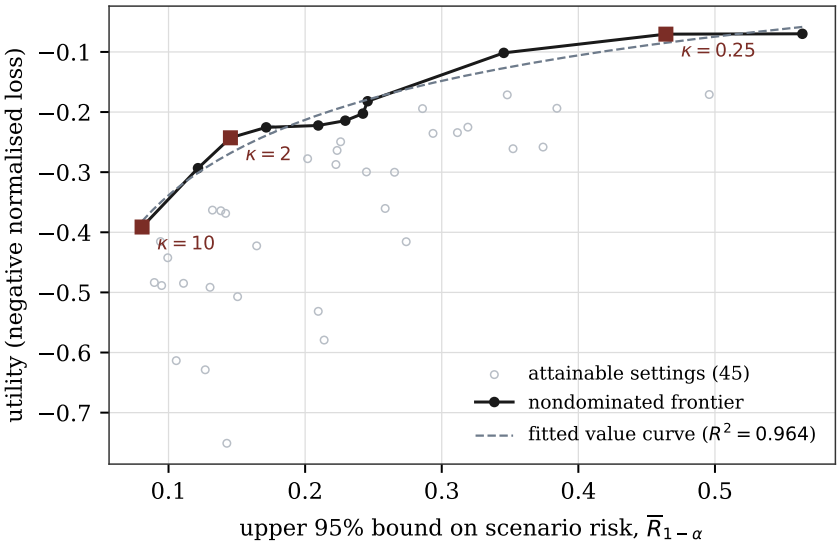


Figure 5: Risk–utility plane for 45 mechanism settings. The risk coordinate is the one-sided 95 per cent bound for a uniformly drawn target, not the point estimate. Squares mark the value-maximising setting for three values of the harm-to-value price ratio κ; the associated risk is the implied tolerance $\tau^\star$ of Proposition 6.

It is steep at the low-risk end: halving the bound from 0.1715 to 0.0806 costs 0.17 of utility, half the range the frontier spans, and cuts the records retained from 100 to 58 per cent.

The attainable frontier is not concave. Successive slopes of the eleven nondominated points run from 6.0 down to 0.006 and not monotonically, which is what a discrete mechanism family produces and what Proposition 8 anticipates. A three-parameter power curve fits it with $R^2 = 0.964$ and is concave over the fitted range; it is used only for the allocation in Section 10.6. Table 5 reports the implied tolerance of Proposition 6, computed as the value-maximising choice over the eleven nondominated settings rather than on the fitted curve, and with the one-sided bound as the risk coordinate as (29) requires.

Two features matter for practice. The implied tolerance is flat over wide ranges, since one setting is optimal from $\kappa = 5$ to $\kappa = 10$, and it jumps between ranges: doubling the price of harm from $\kappa = 0.5$ to $\kappa = 1$ cuts the tolerance from 0.345 to 0.145 and moves the release from an unsuppressed file with twelve occupation groups to one with three groups and a minimum cell count of five. The jump is a property of the discrete mechanism family, not of the argument, and it has a practical reading: over most of the range an institution cannot fine-tune its tolerance, and near a breakpoint a small revision of the price of harm changes the release.

10.6 A programme of four releases

Four releases are built from the same sample. Release A keys on district, sex and five-year age band; release B on district, sex and occupation; release C on district, ten-year age band and six occupation groups; release D on sex, ten-year age band and occupation. Each applies a minimum cell count of three, so each is 3-anonymous and, by construction, no release contains a single unique record. Their marginal risks, computed unconditionally so that a target suppressed from a release contributes zero, are 0.0768, 0.0711, 0.0884 and 0.0168.

Table 5: Tolerance implied by the harm-to-value price ratio, computed by maximising $U - \kappa\overline{R}_{0.95}$ over the nondominated settings. The selected setting reports age band width in years, occupation groups and minimum cell count.

$\kappa = c_h/\lambda$	Implied $\tau^\star$	Selected setting	Records kept
0.25	0.464	10-year bands, 12 groups, $k_{\min} = 1$	1.00
0.50	0.345	20-year bands, 12 groups, $k_{\min} = 1$	1.00
1.00	0.145	20-year bands, 3 groups, $k_{\min} = 5$	0.94
2.00	0.145	20-year bands, 3 groups, $k_{\min} = 5$	0.94
5.00	0.081	20-year bands, 3 groups, $k_{\min} = 20$	0.58
10.00	0.081	20-year bands, 3 groups, $k_{\min} = 20$	0.58

Table 6: Four 3-anonymous releases from one sample, 20,000 target replicates. No single release contains a unique record. Marginal risks are unconditional; a target suppressed from a release contributes zero to that release and does not narrow the intersection. The pooled adversary intersects the candidate sets of releases 1 to m and selects uniformly from the intersection.

	Records released	Marginal risk	Unique share
Release A (district, sex, fine age)	9,648	0.0768	0.000
Release B (district, sex, occupation)	9,709	0.0711	0.000
Release C (district, ten-year age, six groups)	9,257	0.0884	0.000
Release D (sex, ten-year age, occupation)	10,035	0.0168	0.000
Programme-level quantities			
Lower bound $\max_i R_i$		0.0884	
Upper bound $\min(1, \sum_i R_i)$		0.2530	
Value under independence		0.2313	
Pooled attack, $m = 2$		0.4114	0.208
Pooled attack, $m = 3$		0.4279	0.222
Pooled attack, $m = 4$		0.4581	0.260

Table 6 and Figure 6 report what the programme permits. The bounds implied by the marginals are $[0.0884, 0.2530]$ and the independence value is 0.2313. An adversary that intersects the candidate sets of the first two releases succeeds with probability 0.4114, which is 1.6 times the upper bound implied by all four marginals. With all four releases the pooled risk is 0.4581, which is 1.8 times that bound, and 26.0 per cent of targets are uniquely identified by the intersection although none was unique in any single release.

Two remarks. First, the effect is not driven by a large number of releases. Two are enough, and the third and fourth add little because the first two already carry most of the separating information. Second, ranking releases by their own risk figures does not rank their contribution to the programme. Release D has the smallest marginal risk, 0.0168, yet adding it to the pool raises the risk by 0.030, while release C, whose marginal risk is the largest at 0.0884, adds 0.017. Both observations follow from Proposition 10 and neither is visible in a marginal assessment.

Applying Proposition 11 to the union event, with value weights $1, 0.7, 0.5, 0.3$ and $\tau_{\text{prog}} = 0.50$, the optimal allocation is $(0.186, 0.139, 0.106, 0.070)$ against the equal split of 0.125 each, and cuts the total loss by 4.5 per cent. A constrained numerical optimum agrees with the closed form (39) to five decimal places, which is one check on both. With the fitted $\gamma = -0.23$ the

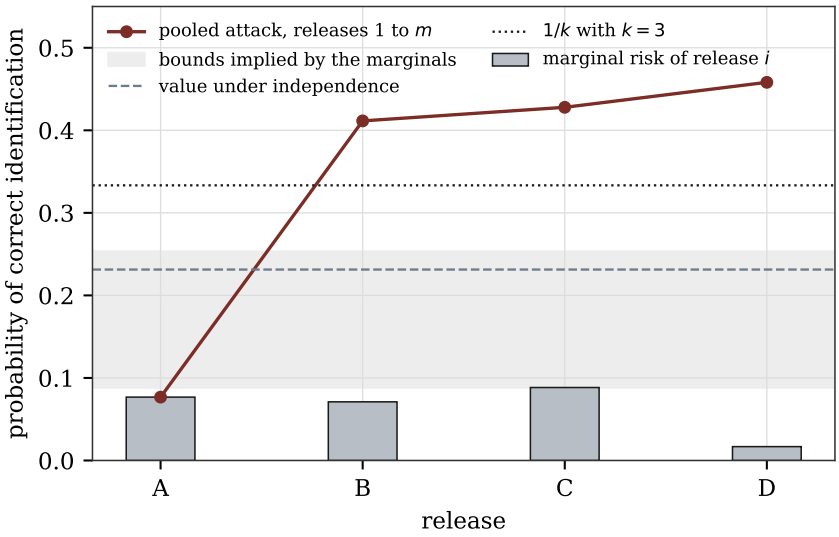


Figure 6: Marginal risks of four 3-anonymous releases and the risk of an adversary that pools them. The shaded band is the interval implied by the marginals under Proposition 9. The pooled attack leaves the band at $m = 2$.

Table 7: Does the pooling gap survive at other register sizes? The same four releases, the same keys and the same minimum cell count, rebuilt on registers of four sizes with the sampling fraction held fixed. Larger registers are safer release by release, because the key classes grow. The pooled attack falls more slowly, so the ratio of the pooled risk to the bound the marginals imply rises with size rather than falling. The gap the paper is about is not an artefact of the size chosen for the main study.

Population	Sample	Largest marginal	Pooled, 2 releases	Pooled, 4	Boole bound	Ratio
50,000	2,362	0.113	0.454	0.623	0.398	1.57
100,000	5,056	0.102	0.458	0.561	0.328	1.71
200,000	10,035	0.088	0.411	0.458	0.253	1.81
400,000	20,079	0.062	0.320	0.332	0.162	2.05

exponent in (39) is 0.81, so the allocation is flatter than the value weights. This is a real gain and a small one, and it is available only for the union event.

10.7 Does the gap survive at other register sizes?

A synthetic study invites the question whether its findings are an artefact of the size chosen. National registers hold millions of records, not the 200,000 of the main study. Table 7 rebuilds the same four releases, with the same keys and the same minimum cell count, on registers of 4 sizes with the sampling fraction held fixed.

Two things move in opposite directions, and the second is the one that matters. Release by release, larger registers are safer: the largest marginal risk falls from 0.113 at 50,000 units to 0.062 at 400,000, because the key classes grow with the population. The pooled attack also weakens, but more slowly, so the ratio of the pooled risk to the bound the marginals imply rises from 1.57 to 2.05 across the same range. Size does not dissolve the problem. It makes the marginals a worse guide to the programme, which is the opposite of the reassurance a large

Table 8: Independent simulated attacks required to certify $\overline{R}_{1-\alpha} \leq \tau$ with no observed success, from (22).

τ	B at $\alpha = 0.05$	B at $\alpha = 0.01$
0.05	59	90
0.01	299	459
10^{-3}	2,995	4,603
10^{-4}	29,956	46,050
10^{-5}	299,572	460,515

register is usually taken to provide. The mechanism is straightforward: growth adds records to every key class, which protects each release, and it does nothing to stop the intersection of four candidate sets from closing on a single record.

The extrapolation beyond the largest size tested is not established here, and a register of several million would need its own run, which the pipeline supports by changing one number in the configuration.

10.8 The cost of certifying a small tolerance

Table 8 applies Proposition 3. The table answers a question that arises whenever an institution proposes a very small tolerance: what evidence would establish it. At $\tau = 10^{-5}$ and $\alpha = 0.01$ the answer is 460,515 independent attack simulations with no success. If the institution cannot run them, and if its replicates share an auxiliary file so that they are not independent, then the tolerance is not supported by simulation and the record should say so.

10.9 How much the binomial interval misses

Section 4 separates simulation error from the variance carried by the sample and the fitted inputs. The separation is testable, and the nested design of Appendix B measures it. 120 outer replicates redraw the 5 per cent sample from the same population, apply the same mechanism, and run 500 attacks each.

The within-replicate standard deviation is 0.0112, and the between-replicate standard deviation comes out at 0.0000 after the correction of Appendix B: on this design the sampling term is not detectable, and the standard error of the mean is 0.0010 whether it is included or not. The figure a decision needs is not that point estimate but its interval, and the interval for the share of variance the design contributes runs from 0 to 0.19. So the honest reading is that the binomial expression is adequate here, and that the data would not have distinguished a design contribution of up to a fifth from none at all. Reporting 0 without the interval would repeat, inside the section on uncertainty accounting, the mistake the paper objects to elsewhere.

Three features of this design keep the term small: the population is fixed and large, the sampling fraction is modest, and the adversary's model is not refitted between replicates. None is guaranteed in practice. The point is the accounting rather than this particular split, and the accounting has a practical edge: increasing B shrinks only the term the binomial expression describes, so an institution that reports a very tight interval from a very large B should say

which of the two terms it drove down.

10.10 Reproducing the study

Everything above comes from one program with a single seed, packaged as Mahomane [2026]. The package keeps four kinds of work apart and labels them: proof checks that verify algebraic consequences of the declared assumptions, simulation checks that estimate attack success and its Monte Carlo error, falsification tests that can reject the model, and the governance gate that applies the tolerance. A run writes the configuration it used, every result, the validation status, the five figures, the LaTeX fragments for the tables in this section, a decision record naming the tolerance, owner, expiry date and adversary, and a SHA-256 hash of each of those files. A failed falsification test produces a non-zero exit status, so a rejected model cannot be presented as governed evidence.

Two details of that arrangement matter for a reader checking this paper. The tables in this section are generated by the run rather than typed, so a table cannot disagree with the experiment that produced it. And the figures quoted in the prose are checked against the run by a script that fails when one of them drifts. The run reported here selects the 20-anonymous setting under a tolerance of 0.12 and returns a pass, with the programme finding recorded beside it.

11 Real data

Everything above is measured on data we generated, which is the only way to check an estimator against a truth. It is not a substitute for running the framework on a file someone actually published. This section does that, on two real public microdata files that ship with the `wooldridge` package: an extract of the US Census 2000 public use sample, 29,501 records with state, public use microdata area, years of schooling, potential experience and log weekly income; and the Boston Fed mortgage application study, 1,986 real applications with area, income, marital status, dependants, occupancy, the lending decision and a record of late payments. Both files are already public and already protected. Nothing confidential is read here, and the question asked is the one an agency asks before publishing a file of that shape: what would a declared adversary achieve against it.

Three things change when the data are real, and each of them is a claim the earlier sections make. The population behind the file is not observed, so the population-frequency measure cannot be computed at all and the journalist measure has no truth to be checked against: a fitted model is the only route to it, which is why Section 16 asks for a frequency-transport test rather than a comparison. The analyses that define utility are given by what the file is used for, not chosen by us. And the risk levels are much higher than in the simulation, so the tolerance question is sharper.

11.1 The census sample

On the full key, state, area, schooling and experience, the 29,501 records occupy 27,390 classes and 86.3 per cent of them are unique. The mean reciprocal frequency is 0.928. This is a

Table 9: US Census 2000 public use sample: the nondominated releases. The file holds 29,501 records, and the columns report the width of the area, schooling and experience bands, the minimum cell count, the share of records kept, the risk under exact matching, the simulated risk of the declared adversary with its one-sided bound, and the utility across the three income analyses. The adversary misreads the key with probability 0.10 and holds a noisy reading of log weekly income.

Area	Schooling	Experience	$k_{\min}$	Kept	Exact matching	Simulated	$\overline{R}_{0.95}$	Utility
100	4	10	10	0.284	0.0615	0.2400	0.2588	-0.310
100	4	10	5	0.566	0.1093	0.3540	0.3748	-0.131
10	4	10	5	0.554	0.1165	0.3553	0.3762	-0.116
100	4	10	1	1.000	0.2965	0.5273	0.5488	-0.000

published file, so the finding is not that someone blundered: it is that the key an outsider can reconstruct almost determines the record, and that any protection claim about it has to come from something other than the key. The key space, counting the areas actually observed inside each state, holds 665,896 cells, and a negative binomial law fitted to the observed counts puts the mean population cell at 0.886 people once empty cells are counted. Under that fitted law a record unique in the file is unique in the population with probability 0.225. There is no truth here to compare that with, which is exactly the position an agency is in.

Table 9 reports the releases on the frontier of 25 settings that vary the area, schooling and experience bands and the minimum cell count. Two features carry over from the simulation and one does not. The declared adversary, who misreads the key with probability 0.10 and holds a noisy reading of income, beats exact matching at every setting, by a factor between 3.0 and 3.9: at the widest bands with a minimum cell count of ten, exact matching gives 0.062 and the simulated attack gives 0.240. The frontier is short, four settings out of 25, and it sits between 0.259 and 0.549, so every release of this file that keeps its analytical value carries a risk an order of magnitude above the tolerances the literature discusses. Moving the price of harm from $\kappa = 0.5$ to $\kappa = 2$ takes the implied tolerance from 0.549 to 0.259 and the records kept from all of them to 28 per cent.

The supremum control is where the real file departs sharply from the simulated one. Take the setting with a minimum cell count of five and examine the 80 records sitting in the smallest classes. Their mean risk is 0.428, and the worst of them is identified with probability 0.98, whose Bonferroni-corrected lower bound is 0.925 against a nominal $1/k = 0.2$. A criterion that promises one in five delivers, for the record it protects least, better than nineteen in twenty. In the simulated register of Section 10.4 the same comparison ran to 0.74 against 0.10; on the real file the covariate does more work, because income varies more inside a class of five real people than the simulation assumed.

Composition behaves as Proposition 10 says it must. Four releases were built from this one file by dropping one key component at a time, each with a minimum cell count of three. Their marginal risks are 0.138, 0.030, 0.118 and 0.096, so the union bound is 0.381. The pooled adversary reaches 0.709, which is 1.86 times that bound, and 60.7 per cent of targets are uniquely identified by the intersection although none of the four releases contains a unique record. The effect is larger on real data than on the synthetic register, where the same comparison gave 1.8

times the bound and 26.0 per cent uniqueness, because real key components are less redundant than generated ones.

11.2 The mortgage applications

The loan file is small and shows the other three risks of Section 12.1. On the key of area, income band, marital status, dependants and occupancy, the 1,986 applications occupy 150 classes, 3.2 per cent of them are unique, and the worst class holds one record. The declared adversary, holding a noisy reading of the loan amount, identifies 0.134 of applications with a one-sided bound of 0.149.

Attribute disclosure is the risk this file makes concrete. 12.2 per cent of applications sit in a class where every application received the same decision, and 0.65 per cent sit in a class where every application was denied, so for those applicants the decision is disclosed by the class rather than by the record. For the record of late payments the shares are 29.6 per cent and 0.35 per cent. Both harmful directions are non-zero here, where the synthetic credit register had none, which is the difference between a base rate chosen for a simulation and one that occurred.

Published totals leak by subtraction as Appendix B describes. Aggregating by area, occupancy and income band gives 48 cells, of which 16, a third of them, contain a single application, so that application's amount is the published total. Those cells hold 2.2 per cent of the total lending in the file. A three-record rule would suppress 42 per cent of the cells.

11.3 What the real files settle and what they do not

Every claim this paper makes on generated data reappears on files that were actually published: the covariate beats exact matching, the supremum control breaks a k-anonymity promise by a wide margin, and pooling defeats the marginal bound. Two of the three are larger on the real files than on the synthetic ones.

Three limits should be read with the numbers. The files are samples whose populations are not observed, so the journalist figure rests on a fitted law that this exercise cannot check, and the utility analyses are computed against the file rather than against the population it came from. The adversary is declared, not observed: the reporting error rate and the precision of the income reading are assumptions, and Section 16 is how an institution would test them. And these are public files, so the exercise measures what a release of this shape discloses, not whether a wrong was done in publishing them.

12 Case studies: what the framework changes

The cases below are not pooled into one empirical estimate. They concern different data products, adversaries and harm classes, and the published evidence does not support a common denominator. Their purpose is sharper: each case identifies which component of the proposed decision record would have made the public claim more precise. Table 10 records that mapping. The first case is treated at length and with measurements, because it is the setting this paper was written for.

Table 10: Case studies mapped to the framework

Case	Release and attack	Mathematical object	Governance lesson
Granular credit register	Loan-level file, aggregates and monthly revisions	Identification, predictive value of an alleged relationship, attribute and residual disclosure, differencing	One lever does not govern five risks; each needs its own target set and tolerance.
Netflix Prize	Sparse ratings linked to outside ratings	Posterior linkage p_{tj} and pooled auxiliary information	"Anonymous" identifiers do not define an adversary or bound posterior success.
US Census 2020	Published tabulations and database reconstruction	Programme composition and formal privacy accounting	Repeated exact statistics require a programme guarantee, not isolated marginal checks.
England and Wales Census 2021	Tables protected by record swapping and cell-key perturbation	Mechanism risk–utility frontier	Operational protection must be paired with an explicit scenario, uncertainty statement and decision owner.
HIPAA de-identification	Safe Harbor or Expert Determination	Rule-based versus scenario-based threshold	A legal route is not itself a numerical probability; Expert Determination makes assumptions and residual risk reviewable.

12.1 A granular credit register

Central banks collect loan-level credit data under statute and publish from it continuously. The euro area collection defines the reporting unit as the instrument and the counterparty, so a single record ties a named borrower to a named lender [ECB, 2022], and the Irving Fisher Committee has documented the same pattern across jurisdictions [IFC, 2016]. National confidentiality policies restrict what may leave the institution, and a published Bank of England policy is one example of the rules a statistics division works under [Bank of England, 2023]. The releases that follow are of three kinds at once: a loan-level file for research use, aggregates by region, sector and lender, and a monthly series with revisions. That combination is what makes a credit register a different problem from a demographic file, and it is the problem this section measures.

Five harms have to be told apart, because a single risk figure covers only one of them. A borrower may be identified. A borrower–lender relationship may be alleged, whether or not the borrower is identified. Arrears, default or forbearance status may be inferred for a class of borrowers without identifying anyone. One lender's contribution may be reconstructed from published totals by another participant. And two consecutive vintages may be differenced. Table 11 reports all five against one lever, a minimum cell count on the borrower key.

The simulated register holds 32,581 relationships between 20,000 borrowers and 60 lenders, with 18 sectors, 12 regions and 4 size classes, so the borrower key has 864 cells, and six monthly vintages with entries and exits. Exposure is lognormal within size class, and arrears occur at 7 per cent. Appendix C gives the details. The lenders hold unequal shares, the largest 23 per

Table 11: Credit register: five risks and one utility under a minimum cell count on the borrower key. The register holds 32,581 relationships between 20,000 borrowers and 60 lenders. Identification is the average over released relationships with its one-sided bound, and again for the largest size class alone. The predictive value is that of an alleged borrower-lender relationship, and the homogeneity share is the proportion of large borrowers whose key class discloses their arrears status. Utility is the negative weighted loss across credit growth, concentration, the arrears ratio and lender-level supervision.

$k_{\min}$	Kept	Identification	$\overline{R}_{0.95}$	Large firms	Relationship PPV	Status disclosed	Utility
1	1.000	0.0695	0.0765	0.185	0.345	0.427	-0.000
5	0.993	0.0722	0.0793	0.204	0.343	0.388	-0.039
10	0.970	0.0653	0.0720	0.141	0.339	0.291	-0.115
20	0.889	0.0583	0.0647	0.089	0.332	0.160	-0.204

cent.

Identification is a subgroup problem. Success can be scored two ways, and the difference matters. Recovering the exact relationship succeeds for 0.0695 of released rows, with a one-sided bound of 0.0765; recovering the right borrower, which is the harm the register creates, succeeds for 0.0925, because a borrower with several lenders can be found through any of them. Both look comfortable as averages. Split by size class the relationship figure is 0.040, 0.082, 0.092 and 0.185, so a large borrower faces more than four times the risk of a micro borrower. This is the target-set point of Section 5: the same file satisfies any tolerance its average figure meets and breaches that tolerance on the largest size class, and that class is the one whose distress moves markets. Raising the minimum cell count to 20 brings the large-borrower figure to 0.089 and keeps 89 per cent of the relationships. That subgroup rests on few targets, so it is reported with its interval: [0.138, 0.240] at $k_{\min} = 1$ and [0.044, 0.158] at $k_{\min} = 20$. The two overlap, and the risk is not monotone across the grid, rising to 0.204 at $k_{\min} = 5$ before falling. A paper arguing that point estimates are not decisions should not make an exception for its own subgroup column: what the evidence supports is a fall of roughly half, not a measured halving, and a lever that is unreliable at its first step.

An alleged relationship is a base-rate statement. Suppose the adversary names, for every borrower in the release, the lender holding the largest share of that borrower's key class. Equation (69) gives the quantity that matters, the probability that an allegation is true. On the simulated register the prevalence of any given borrower–lender pair is 0.0261, the sensitivity of this rule is 0.221 and its specificity is 0.989, and (69) returns 0.345, which is the realised share of correct allegations. The four cells are reported so that a reader can check the identity rather than take the predictive value on trust; their agreement is arithmetic, not evidence. Two readings follow. A specificity of 0.989 sounds protective and is not, because the prevalence is low. And an allegation that is right a third of the time is 1.5 times better than naming the largest lender at random, which is 1.5 times the success rate of always naming the largest lender in the market, and wrong about two borrowers in three. The minimum cell count barely moves it: 0.345 at $k_{\min} = 1$ and 0.332 at $k_{\min} = 20$. The lever that governs identification does not govern relationship disclosure, and an institution that reports only the first has not addressed

the second.

Status can be disclosed without identification. A key class whose members all share an arrears status discloses that status for every member, whatever the class size. For large borrowers 42.7 per cent of released relationships sit in such a class at $k_{\min} = 1$, falling to 16.0 per cent at $k_{\min} = 20$. In this register every homogeneous class is homogeneous in the benign direction, because arrears are rare, so what is disclosed is the absence of arrears. Large borrowers carry the highest homogeneity share because their key classes are the smallest, averaging 10.9 members against 74.9 for micro borrowers. That is a property of the base rate and the class sizes rather than a guarantee, and it is exactly the quantity an agency should measure on its own register, where a small sector–region cell in distress is the case that matters. A minimum cell count does not address the harm; the attribute-level criteria of Machanavajjhala et al. [2007] and Li et al. [2007] do, and they change the risk model rather than the tolerance.

Published totals give contributions away by subtraction. Published totals by region, sector and size class give a linear system, and Appendix B states when a contribution is determined by it. In the simulated register 1.0 per cent of the 817 published cells have a single contributor, so that lender's position is the published total, and a further 2.7 per cent have two, so either participant recovers the other's amount by subtraction. Together that is 3.7 per cent of cells, holding 2.0 per cent of total exposure; the single-contributor cells alone hold 0.44 per cent. The remedy is a rule on the aggregates, not on the microdata: requiring five contributors rather than three suppresses 12.5 per cent of cells instead of 3.7 per cent. The loan-level minimum cell count leaves this risk untouched, which is the third place in this case where one lever fails to govern another risk.

Monthly vintages can be differenced. When a published cell's count moves by one between vintages, the change in the published total is the arriving or departing amount, provided one relationship moved. The lever here is not the loan-level minimum cell count but a minimum on the published count, and it is written $c_{\min}$ to keep the two apart. Across consecutive months 11.0 per cent of comparable cell pairs show a unit change at $c_{\min} = 1$, falling to 5.6 per cent at $c_{\min} = 10$. The inference is not certain, because the adversary does not observe the number of moves, and (70) prices that uncertainty. Which rate goes into it decides the answer. One rate pooled over all cells, 4.7 movements per cell per transition, returns 0.003, which would dismiss the attack. The rate of the cell in question returns 0.61, because the cells that show a unit change are the quiet ones, averaging 2.4 movements against 9.3 across all cells. In this register the realised share is 1.00: every unit change was in fact a single move. The three figures span the whole range, and the reason is that the movement model, not the arithmetic, is doing the work. A differencing figure is therefore a statement about a fitted movement model, and it has to be fitted at the level at which the inference is drawn and checked against realised moves, exactly as Section 16 requires of the linkage model. Reporting the pooled number alone would have understated this register's exposure by three orders of magnitude.

The utility side is not one number either. A credit register is collected for credit growth measurement, concentration analysis, financial stability monitoring and supervision, and a minimum cell count damages them unequally. At $k_{\min} = 20$ the weighted loss is 0.204, of which almost all comes from the sectoral exposure and lender-level supervision terms, at 0.432 and 0.359, while the concentration and arrears-ratio terms lose 0.011 and 0.004. Both level estimates are grossed up by the retained share of records, which is the rule an analyst applies who is told how many records were suppressed and not how much exposure they carried; suppression removes the thin large-borrower cells, so that rule understates the totals and the loss reflects both effects together. Suppression removes whole thin cells, which is where a supervisor looks and where an aggregate ratio does not change. An institution that reports one utility score for the release hides which analysis paid for the protection.

What the record should say. The five risks have different target sets, different harm classes and different owners, so they need five entries in (28) rather than one. No tolerance has been stated for this register, and until one is, neither *pass* nor *fail* means anything under Definition 2. On this register a defensible record would set a subgroup tolerance for identification of large borrowers, which at $\tau = 0.10$ would admit only the $k_{\min} = 20$ setting, a separate tolerance on the predictive value of an alleged relationship, an attribute rule rather than a cell-count rule for status, a contributor rule on the aggregates, and a programme tolerance for the monthly series assessed by the joint simulation of Section 7, since consecutive vintages are exactly the pooled adversary of Proposition 10. No single minimum cell count delivers all five, and Table 11 shows why: as $k_{\min}$ rises from 1 to 20 the large-borrower identification risk falls by about half, the relationship predictive value falls by four per cent, and the reconstruction risk does not move at all.

12.2 Sparse behavioural data: the Netflix Prize

The Netflix Prize data replaced direct identifiers with pseudonymous user numbers, but the released rating vectors were high-dimensional and sparse. Narayanan and Shmatikov [2008] showed that a small amount of auxiliary rating information could link records across datasets. In the notation of Section 3, the released identifier was irrelevant to the attack: Z_t contained a few ratings and approximate dates, L_{tj} measured their agreement with candidate records, and the largest posterior in (11) determined the match. The case therefore illustrates the failure of exchangeability, not merely the presence of sample uniques.

The defensible retrospective test is not whether the file was anonymised. It is: for a prespecified distribution of auxiliary ratings and date error, what is the distribution of $\max_j p_{tj}$ across targets, and what simultaneous upper bound lies below the owned tolerance? A modern replication should vary the number of auxiliary ratings, their popularity, the date tolerance and the abstention threshold. Those variables define a family of attacks; choosing only an easy one after seeing the outcome would not validate the release.

12.3 Repeated official statistics: the 2020 US Census

The US Census Bureau adopted a disclosure avoidance system built on differential privacy after demonstrating that many published tabulations can support database reconstruction [Garfinkel et al., 2019, Abowd et al., 2022, 2023]. This is the institutional case closest to Proposition 10. The threat does not arise because one table has a large marginal linkage risk. It arises because many accurate, overlapping constraints describe the same underlying population.

The framework makes two distinctions. First, reconstruction, membership inference and correct named-person linkage are different events and must not be reported under one risk label. Second, a formal privacy accountant governs the programme mechanism, whereas $\sum_i R_i$ governs only a union of declared single-release attacks. The policy debate over accuracy remains a risk–utility choice, but the protection claim and its unit of composition are now explicit. The appropriate empirical study would compare redistricting, small-area and demographic estimands under alternative privacy-loss budgets, then report subgroup-specific utility losses beside the programme guarantee.

12.4 Layered operational protection: Census 2021 in England and Wales

The Office for National Statistics describes protection of Census 2021 outputs through record swapping and the cell-key method, together with checks against disclosure in tables [ONS, 2022]. This is not evidence that the release has a universal linkage probability, nor need it be. It is evidence of a layered mechanism M_θ whose tuning changes both tabular accuracy and the information available to an intruder.

For governance, what is missing is a declared attack experiment. An evaluation could draw targets from separately stated population groups, expose the attacker to realistic public registers, reproduce the cell-key and swapping randomness, and calculate $\overline{R}_{1-\alpha}$ for each protected group. The mechanism parameters, the seeds policy, the table family and the expiry date would then be part of G in (28). That turns a process description into a reproducible release decision, without implying that the mechanism offers differential privacy unless it does.

12.5 Health data: HIPAA Safe Harbor and Expert Determination

US health-data guidance recognises two routes for de-identification: removal of specified identifiers under Safe Harbor, and Expert Determination that the risk of identification is very small, using generally accepted statistical and scientific principles [HHS, 2012]. The contrast maps directly to this paper. Safe Harbor constrains the released variables; Expert Determination is a scenario-conditioned judgement about residual risk. Neither phrase alone fixes T, a, h, τ or α.

A rigorous determination would therefore file the population and target set, the accessible auxiliary sources, the linkage rule, the harm class, the estimate, the uncertainty bound and the review horizon. Where data are released repeatedly to the same recipient, the determination must count the accumulated products as part of what the adversary holds. This does not replace legal judgement. It makes the empirical premises of that judgement capable of expiry, challenge and replication.

12.6 A common case-study protocol

Across the five settings, a comparable prospective study has seven steps:

1. freeze the release mechanism, and define neighbouring or alternative confidential states where a formal guarantee is claimed;
2. specify the target population, the auxiliary information and the abstention rule;
3. separate linkage, membership, attribute and reconstruction success events;
4. simulate or execute attacks with the mechanism randomness and the shared data dependencies preserved;
5. estimate target and subgroup risks with simultaneous uncertainty bounds;
6. measure utility on prespecified authorised analyses, including subgroup error rather than only an aggregate score; and
7. assess the complete release programme, then file the governed tuple G.

The resulting studies would be comparable in design while keeping different numerical scales where the estimands genuinely differ.

13 An adversarial evaluation of a credit-release system

Section 12.1 measured five risks on a granular credit register against one protection lever. This section does what an evaluation of a privacy-critical system asks for instead. It fixes an adversary, fixes a set of attacks, and runs them against the protection policies that statistical offices and supervisors actually operate, on one register, with one body of auxiliary information. The results are reported the way an attack is judged: how often it succeeds, how often it is right when it chooses to act, what it must know before it can act at all, and how it degrades when its own model of the data is wrong.

The purpose is not to show that a new mechanism is better. It is to show that the quantity these policies are calibrated against, the risk of one release, does not govern the risk of the programme they are part of. Proposition 10 says no bound in terms of the marginals exists. The evaluation here measures the size of that gap on a release system of the kind supervisors run every month, and prices it in records.

13.1 Threat model

The adversary targets one borrower and holds four pieces of auxiliary information about it: sector, region, size class, and one reading of log exposure per published vintage. Each reading is corrupted by independent Gaussian noise. The key fields are held exactly, which is the pessimistic case for the institution and the natural one for a credit register, where a firm's sector, region and size class are matters of public record.

The attack blocks on region, restricts to released loans whose key agrees with the target's, scores each candidate by the density of the exposure reading under the adversary's own error model, normalises the scores to a posterior over candidates, and names the maximum. It names a record only when that posterior passes one half, so the adversary may abstain, and precision is a quantity with content rather than an artefact of forcing a guess. Pooling multiplies the evidence across the vintages in which the target appears. Nothing else changes between the per-release and the pooled attack: the same policy, the same target draw, the same readings. Each entry in Table 12 rests on 4,000 simulated attacks against independently drawn targets.

Three further attacks run against the same releases. The relationship attack names, for each released borrower, the lender holding the largest share of that borrower's key class, and is scored by the share of allegations that are true. The attribute attack reads arrears status off a key class whose members agree, and is scored by accuracy on the classes it is confident about. The differencing attack compares two vintages of the same cell and claims a single arrival or departure whenever the count moves by one, and is scored by how often exactly one loan moved.

13.2 The policies compared

7 policies enter the comparison, and each of them is in operational use somewhere:

- no suppression, the isolated-release baseline;
- the minimum-contributor rule, which withholds a cell unless at least three lenders contribute to it, the rule most commonly applied to supervisory aggregates;
- the same rule with complementary suppression, which also withholds the smallest surviving cell in each region so that a suppressed cell cannot be recovered from a published row total;
- k-anonymity on the borrower key at $k = 5$ and $k = 10$;
- release of counts alone under Laplace noise, at two privacy parameters, which is the compositional mechanism a formal-privacy programme would use.

Utility is the same weighted loss across credit growth, concentration, the arrears ratio and lender-level supervision that Section 12.1 used, computed on what each policy actually publishes. The noisy-counts policy carries a fixed penalty for the analyses that loan-level work supports and counts do not.

13.3 What the attacks achieve

Table 12 reports the matrix. Four findings stand out.

First, every suppression policy reduces the per-release risk and none of them removes it. With no suppression the adversary names the right loan for 0.079 of targets. The minimum-contributor rule brings that to 0.067, k-anonymity at $k = 5$ to 0.062, and at $k = 10$ to 0.055. Pooled over the programme, the two rules stand level at 0.209, although they differ per release. The reduction within a release is real and it is modest, because the binding constraint is not

Table 12: Attack outcomes by protection policy, on the same credit register and the same adversary. The release and pooled columns give the share of targets the maximum-posterior linkage names correctly, for one release and pooled over the 6 vintages. Precision is measured over the attempts the adversary acts on, those whose posterior passes 0.5, and the column headed acts is how often it acts at all. The last column is the Bonferroni-corrected upper bound on the worst size class. Utility is the negative weighted loss on the four analyses the register is collected for.

Policy	Kept	Utility	Release	Pooled	Precision	Acts	Worst
no suppression	32,581	-0.000	0.079	0.209	0.774	0.015	0.122
three contributors	32,513	-0.013	0.067	0.209	0.674	0.011	0.065
complementary suppression	32,467	-0.021	0.069	0.225	0.762	0.011	0.076
k-anonymity, $k = 5$	32,351	-0.039	0.062	0.224	0.761	0.011	0.061
k-anonymity, $k = 10$	31,615	-0.115	0.055	0.194	0.536	0.007	0.036
noisy counts, $\varepsilon = 1$	–	-0.400	0.000	0.000	–	0.000	–
noisy counts, $\varepsilon = 5$	–	-0.360	0.005	0.005	1.000	0.005	–

cell size but the exposure reading, which separates candidates inside a cell however large the cell is.

Second, the same policies do far less against the pooled attack. Under $k = 5$ the per-release figure of 0.062 becomes 0.224 once the adversary combines the vintages, and under $k = 10$ it becomes 0.194. The largest gap in the table, 0.162, belongs to k-anonymity, $k = 5$. A policy that looks adequate on the release it was written for is not adequate on the programme it is part of, and the ordering of policies by per-release risk is not the ordering by pooled risk.

Third, the adversary is accurate when it acts. Under $k = 5$ it passes the decision threshold on 0.011 of its attempts and is right on 0.761 of those. Low recall with high precision is the operationally dangerous combination: an attacker who needs one confirmed identification, not many, gets it cheaply, and an evaluation that reports only the average success rate hides this.

Fourth, the average conceals a subgroup. The worst size class under $k = 5$ is identified at 0.022, with a Bonferroni-corrected upper bound of 0.061 over the classes examined. Large borrowers are few, so their key classes are small and their exposures are far apart, and both effects push the same way.

The other three attacks are largely indifferent to the lever. Under $k = 5$ an alleged borrower-lender relationship is true 0.343 of the time, a confident reading of arrears status is right 0.944 of the time, and a unit change between vintages is a single loan movement 1.000 of the time. Suppression on the borrower key is aimed at identification, and it does not touch the disclosures that do not require it.

Noisy counts behave differently, as they should. At $\varepsilon = 5$ the adversary identifies 1.000 of the unique cells and is right 1.000 of the time when it claims one, at a utility of -0.360 against -0.039 for $k = 5$. The mechanism moves the analysis onto a different footing, described in Section 9, and the price is the loan-level work that supervision needs.

13.4 Per-release calibration against programme calibration

The headline number is a governance one. Fix the tolerance at 0.12 and ask which minimum cell count is the least restrictive that meets it. Calibrated on one release, the answer is $k = 1$,

Table 13: What the attack needs, and what happens when the adversary's model is wrong. The upper block withdraws auxiliary information from the adversary one field at a time; the lower block keeps all four fields and misstates the dispersion of its exposure reading. Calibration error is the gap between the average posterior the adversary reports and the share of its guesses that are right. Both blocks use k-anonymity at $k = 5$.

Adversary holds	Fields	Risk	Precision	Calibration error
all four fields	4	0.072	0.800	0.002
no exposure reading	3	0.026	–	0.002
no size class	3	0.041	0.900	0.003
region and exposure only	2	0.003	0.000	0.000
assumed $\sigma \times 0.5$	4	0.066	0.341	0.068
assumed $\sigma \times 1$	4	0.066	0.632	0.001
assumed $\sigma \times 2$	4	0.066	–	0.032

which keeps 32,581 relationships. Calibrated on the pooled attack over the programme, the answer is $k = 50$, which keeps 20,389. The per-release rule certifies a release programme whose realised risk is above the tolerance the institution set, and it does so while appearing to pass its own test at every vintage.

This is the practical content of Proposition 10. The proposition denies that any function of the marginals can bound the pooled risk. The evaluation shows what that denial costs on an ordinary register: not a contrived construction, but a monthly credit collection with a lever every supervisor recognises. The programme-level certification of Section 7 is what closes the gap, and Table 12 prices it: noisy counts, $\varepsilon = 5$ is the policy in the comparison set with the highest utility whose pooled risk meets the tolerance.

13.5 What the attack needs

Table 13 withdraws the adversary's auxiliary information one field at a time, under k-anonymity at $k = 5$. The full adversary reaches 0.072. Without the exposure reading it reaches 0.026, and it never passes the decision threshold at all, because a released cell has at least five members and nothing separates them. Holding only region and the exposure reading it reaches 0.003.

The reading of a continuous variable, not the categorical key, is what makes the attack work. That has a direct operational consequence, and it is not the one suppression rules assume. Protection effort spent on the key buys less than protection effort spent on the published continuous variable, and an institution that knows what its adversary can read about exposures knows most of what it needs to know about its own risk. It also gives a cost measure: the attack requires one noisy reading per target per vintage, which for a credit register means a data source that tracks firm borrowing, and the evaluation says how much accuracy that source needs to buy.

13.6 When the adversary's model is wrong

The lower block of Table 13 keeps all four fields and misstates the dispersion of the exposure reading in the adversary's own likelihood. The success rate barely moves, because the maximum-posterior choice is invariant to a monotone rescaling of the same density over the

same candidates. What moves is calibration. With the dispersion stated correctly the gap between the posterior the adversary reports and the share of its guesses that are right is 0.001; misstated, it reaches 0.068.

This matters for the institution, not for the attacker. An attacker who is overconfident still identifies the same records. An institution that estimates risk under a misspecified adversary model reports a number whose stated confidence is wrong, and the falsification tests of Section 16 are the check that catches it.

13.7 What this evaluation does not settle

The register is synthetic. Its structure follows a supervisory collection, with skewed lender shares, several relationships per large borrower and monthly entry and exit, but the parameters are chosen, not estimated from a real credit register, which cannot be released. The results are therefore claims about a class of release systems with these structural features, and the way to refute them is to run the same evaluation inside an institution that holds the real file, which the pipeline supports without modification.

The adversary is one adversary. A better one exists: it could use the arrears status, the lender identity, or a longer history. The evaluation is a lower bound on what is possible, and its value is comparative, since every policy faces the same attacker.

Finally, the evaluation compares policies as they are usually configured, not as they could be. A minimum-contributor rule applied to the pooled programme rather than to each vintage would do better than the row for it here, and that is the point: the same lever, calibrated on the right quantity, is what changes the outcome.

14 A multidimensional trust position with a neural proposal model

The word "trust" requires an operational definition. For a proposed mechanism θ, let $j = (a, h, g, v)$ index an adversary, harm class, target group and release vintage, including a separate index for an attacker pooling vintages. The finite, declared index set is $\mathcal{J}$, of cardinality n. A scenario j fixes the target sampling frame, attack algorithm, access to auxiliary information, release programme and binary definition of success. Conditional on a fixed proposed release and these specifications, let $P_{\theta j}$ be the law of one fresh attack experiment and define the actual risk $r_j(\theta) = \mathbb{E}_{P_{\theta j}} S_j$, where $S_j \in \{0, 1\}$ records success. The tolerance $\tau_j \in (0, 1)$ is fixed before the final audit. A population average and a particular target require distinct scenario definitions. Trust cannot be inferred from the mere existence of a neural score. We call the *trust position* the vector of independently audited margins

$$\mathbf{z}(\theta) = \left(1 - \frac{\overline{r}_j(\theta)}{\tau_j}\right)_{j \in \mathcal{J}} \in \mathbb{R}^n, \qquad T(\theta) = \min_{j \in \mathcal{J}} z_j(\theta). \tag{49}$$

Here $\overline{r}_j$ is a simultaneous upper confidence bound, with its sampling unit and adversary written into the assessment. A nonnegative T means every declared coordinate passes. The complete vector must accompany the scalar: the minimum alone conceals which group or programme

attack constrains the release. Utility $U(\theta)$ is kept as a separate coordinate, since a release can have an excellent trust position while carrying little public value. Missing or unaudited coordinates receive $\overline{r}_j = 1$ and therefore fail whenever $\tau_j < 1$. This gives a literal starting point with no inherited claim of trust: collect independent evidence, tighten the bounds, improve the mechanism, and recertify. It does not promise that evidence will rescue an unsafe release.

Figure 7 depicts the coordinatewise decision. Each entry is a separately audited margin, and one negative entry forces the minimum below zero even if the other coordinates pass.

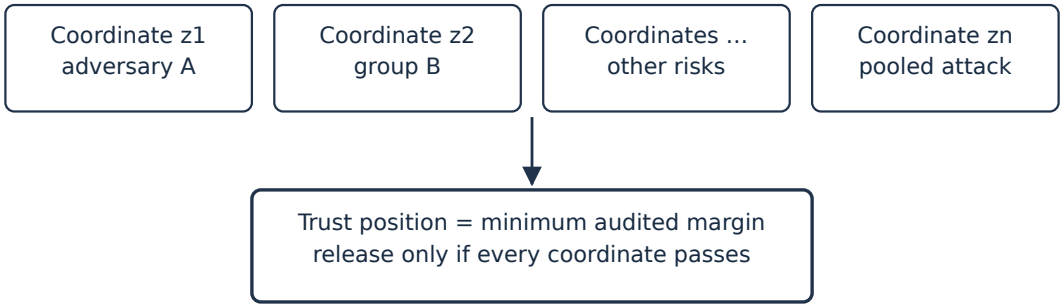


Figure 7: The n-dimensional trust position uses independently bounded risk coordinates $z_j = 1 - \overline{r}_j/\tau_j$. The labels illustrate declared coordinates, not numerical results.

14.1 The neural model and what its output means

Encode a proposed release and its environment as $x(\theta) \in \mathbb{R}^d$: mechanism parameters, sample fraction, key entropy and sparsity, auxiliary information quality, number and overlap of prior releases, access controls, population shifts and target-group indicators. All features must be available *before* certification and their provenance recorded. A transparent baseline is a regularised logistic model; our nonlinear proposal model is

$$H_0 = x, \qquad H_{\ell+1} = \rho(H_\ell W_\ell + b_\ell), \quad 0 \le \ell < L, \tag{50}$$

$$\widehat{\boldsymbol{r}}_\phi(x) = \sigma(H_L W_{\text{out}} + b_{\text{out}}) \in (0,1)^n, \quad \mathcal{L}(\phi) = -\sum_{i,j} w_{ij}[y_{ij} \log \widehat{r}_{ij} + (1 - y_{ij}) \log(1 - \widehat{r}_{ij})]. \tag{51}$$

The label y_{ij} is the *observed success of an attack* on a separate scenario draw, with a zero label for a failed or abstaining attack as specified in the threat model. Unevaluated pairs have $w_{ij} = 0$; repeated attacks sharing one auxiliary file are clustered by file rather than treated as independent. The sigmoid outputs are predicted attack frequencies, not probabilities of institutional trust, and no ordering of them across harm classes is implied. Train on past mechanism–adversary scenarios, hold out whole populations, release programmes and vintages, and keep the final audit draws out of all training, tuning, calibration and feature selection. Compare log loss, Brier score and reliability diagrams with the logistic and frequency baselines, both overall and by group [Guo et al., 2017]. A calibration plot is a diagnostic, never a certificate for the highest-risk individual.

The neural model proposes and ranks a *finite* shortlist Θ_0 by estimated risk and public utility. For each candidate and j, run $B_{\theta j}$ fresh, conditionally identically distributed Bernoulli

attacks and count $K_{\theta j}$ successes. Fix the sample sizes before observing final outcomes. With $M = |\Theta_0|n$ prespecified inspected coordinates, define the exact one-sided bound

$$\overline{r}_{\theta j} = \begin{cases} 1, & K_{\theta j} = B_{\theta j}, \\ F^{-1}_{\mathrm{Beta}(K_{\theta j}+1,\, B_{\theta j}-K_{\theta j})}(1-\alpha/M), & K_{\theta j} < B_{\theta j}. \end{cases} \tag{52}$$

Certification uses (49) and (52), never the neural output. The final choice may maximise U among certified candidates, provided the shortlist was fixed without looking at these final audit draws.

Figure 8 makes the separation between development and final evidence explicit. The arrows into the final gate carry audit bounds and tolerances; the network contributes only the fixed shortlist.

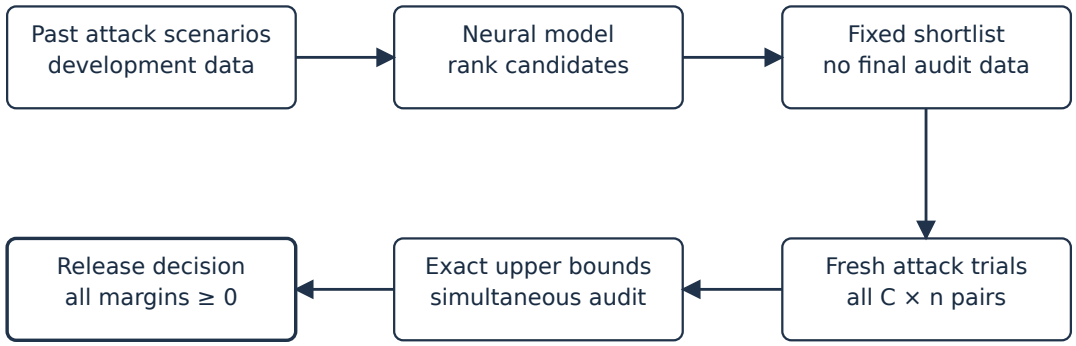


Figure 8: The neural network ranks candidates; independent attack trials and exact simultaneous upper bounds decide. $C = |\Theta_0|$ and n is the number of risk coordinates.

Proposition 15 (Selection-safe finite-family certification)**.** *Condition on development information $\mathcal{D}$ and its resulting finite shortlist, candidate releases, scenarios and sample sizes. Suppose each final audit count is* $\mathrm{Binom}(B_{\theta j}, r_j(\theta))$ *conditional on $\mathcal{D}$; auxiliary information sampled afresh is included in this experiment, whereas fixed auxiliary information is conditioned upon. Then with probability at least $1-\alpha$, simultaneously for every inspected pair, $r_j(\theta) \le \overline{r}_{\theta j}$. Consequently the rule selecting any candidate with $T(\theta) \ge 0$ has probability at most α of certifying a candidate whose risk exceeds a declared tolerance in any coordinate.*

Proof. For one pair the event that the exact upper bound falls below the true risk has probability at most α/M by inversion of the binomial lower-tail test (Appendix D). A union bound over the M pairs gives the simultaneous statement, without requiring independence between pairs. On this event every certified candidate satisfies $r_j(\theta) \le \tau_j$ for all j, regardless of which certified candidate the final rule selects. Taking expectation over $\mathcal{D}$ gives the unconditional guarantee. □

If the release, attack, shortlist or group definitions change in response to final outcomes, the conditional argument no longer applies. Freeze and restart with untouched audit draws, or replace the fixed-sample bound by the anytime procedure of Section 8 with an explicit

multiplicity allocation. Binomial sampling covers attack-draw variability conditional on a fixed population and mechanism. It does not cover uncertain input data, unknown auxiliary sources, a stronger future attacker or distribution shift. Those are separate scenario coordinates or require a stated outer uncertainty analysis.

14.2 What changes as the dimension grows

The number d of input features and the number n of risk coordinates are different. For widths $q_0 = d, q_1, \ldots, q_L, q_{L+1} = n$, a dense forward pass costs $O(\sum_{\ell=0}^{L} q_\ell q_{\ell+1})$ operations per scenario and stores $O(\sum q_\ell q_{\ell+1})$ parameters; a training epoch multiplies the cost by the number of labelled scenarios. Sparse embeddings or shared encoders can reduce cost when many groups share structure, but cannot supply evidence in an unseen group. A full final audit over C candidates and n coordinates uses Cn confidence bounds and, at zero observed successes in every coordinate, needs at least

$$B \geq \left\lceil \frac{\log(\alpha/(Cn))}{\log(1-\tau_{\min})} \right\rceil \tag{53}$$

independent trials per coordinate to certify a common minimum tolerance $\tau_{\min}$. This is necessary and sufficient when that coordinate has zero successes; with successes, more trials can be needed. Thus computational amortisation by a network does not erase the statistical cost of many simultaneous guarantees. For an L-Lipschitz risk function on $[0,1]^d$, even a direct grid at accuracy ϵ uses order $(L/\epsilon)^d$ cells. Without such regularity, finite observations alone cannot guarantee uniform pointwise accuracy. Report learning curves against both d and n; do not describe merely adding hidden units as solving this problem.

For stress testing, if the full law $Q_{\theta j}$ of the deployed attack experiment is known to satisfy $\mathrm{TV}(P_{\theta j}, Q_{\theta j}) \leq \delta_j$, bounded attack success gives $r_{Q_{\theta j}}(\theta) \leq r_{P_{\theta j}}(\theta) + \delta_j$. A declared and justified bound can therefore replace $\overline{r}_j$ by $\min(1, \overline{r}_j + \delta_j)$. Estimating δ_j from a neural embedding alone does not justify that guarantee. Out-of-scope feature values, new linkage data and an unbounded shift require refusal and a new assessment.

The testable methodological contribution is the combination of a vector release position, a neural search model, and an independent simultaneous audit under adaptive choice of the final candidate. Neural prediction, binomial bounds and neural calibration individually have extensive precedents [Bates et al., 2021, Guo et al., 2017]; the claim here is about their stated division of labour for a release programme, not an untested claim of superior empirical performance. The accompanying `privacy_risk.neural_trust` module includes a seeded synthetic demonstration of the search and audit interface. It cannot validate the existing tables in this paper. An empirical extension must preregister targets and populations, compare the network with the logistic and frequency baselines at fixed audit budgets, report selective utility and worst-group failures, and rerun every pooled attack on held-out real programmes.

15 SENTRY-R: scenario-enriched neural trust for releases

We name the implementable extension *SENTRY-R*, shorthand for scenario-enriched neural trust in release decisions. This names the specified workflow, not a claim that its components are

individually new or that a fitted network has improved real release decisions. Its separation of three data streams is central: *untrusted scenario proposals*, *measured development attacks*, and *untouched final audits*. The implementation supports any finite input dimension d and any finite number n of declared risk coordinates. A deeper network predicts attack frequencies for search; it does not define trust.

15.1 LLM-assisted scenario expansion and evidence provenance

Let $\mathcal{F}_t$ denote the information available before development round t, including past mechanisms, validated scenario specifications and measured attacks. A language model proposes descriptions and numeric scenario features $Q_t = G_t(\mathcal{F}_t, \xi_t)$. A deterministic validator checks the predeclared feature names and bounds, scenario identifier, and source metadata. It does not treat free-form text or a language-model risk estimate as an observed success. A separate attack evaluator then draws Z_{ti} under the accepted scenario and records $K_{tj} = \sum_{i=1}^{B_{tj}} s_j(Z_{ti})$, together with its experiment identifier and sampling unit. The retraining history grows by

$$\mathcal{F}_{t+1} = \mathcal{F}_t \vee \sigma\big(Q_t, \mathrm{valid}(Q_t), \{K_{tj}, B_{tj}, \mathrm{provenance}_{tj}\}_{j\in\mathcal{J}}\big)\,. \tag{54}$$

The language model can change which *development* experiment is run next; it does not directly change a network weight or label. Source identifiers, scenario hashes and an explicit partition label make the update inspectable. Treating repeated targets or auxiliary files as independent Bernoulli trials would break the specified sampling model. Human review is still required to establish whether a proposed adversary, harm definition and auxiliary source are plausible and permissible.

Let $\mathcal{D} = \mathcal{F}_{t_*}$ be all such development information at a chosen stopping time t_*, including model weights and hyperparameter selection. A separate validation set holds out entire release programmes; final audit draws are excluded from $\mathcal{D}$. Conditional on $\mathcal{D}$, freeze the coordinate definitions, policy tolerances, C shortlisted candidates and their planned sample sizes. The Proposition 15 guarantee then applies even if the *development* sequence used an LLM adaptively, provided the conditional binomial model for the untouched final draws holds. A final audit reused as development data, or an LLM proposal generated from its outcomes, invalidates that conditioning. Figure 9 shows the implemented boundary.

15.2 A deep model trained on counted attack outcomes

The `privacy_risk.sentry_r` module provides a depth-L fully connected ReLU model with common width q, including the depth-zero logistic baseline. Training features $x_i \in \mathbb{R}^d$ are standardised using *training* means and scales only. With $h_i^0 = x_i$, for $\ell = 0, \ldots, L-1$ define

$$h_i^{\ell+1} = \rho(h_i^\ell W_\ell + b_\ell), \quad a_{ij} = \big(h_i^L W_{\mathrm{o}} + b_{\mathrm{o}}\big)_j\,, \quad p_{ij} = \sigma(a_{ij}), \quad \rho(z) = \max\{0, z\}. \tag{55}$$

For each observed development pair (i, j) let $K_{ij} \in \{0, \ldots, B_{ij}\}$ be the count of successes in $B_{ij} \geq 1$ trials. The implementation minimises the binomial negative log likelihood, without its



Figure 9: SENTRY-R development and decision boundary. The LLM's scenario proposal is checked against a fixed schema; only measured development attacks enter the deep model. Its ranking freezes a shortlist before a fresh, independent final audit.

constant combinatorial term, and a ridge term:

$$\mathcal{L}(\phi) = -\frac{1}{B_\bullet}\sum_{i,j}\left[K_{ij}\log p_{ij} + (B_{ij}-K_{ij})\log(1-p_{ij})\right] + \frac{\lambda}{2}\sum_{\ell=0}^{L}\|W_\ell\|_F^2, \quad B_\bullet = \sum_{i,j} B_{ij}. \tag{56}$$

Here $W_L = W_{\mathrm{o}}$; a weight attached to trial count is appropriate only if the count's sampling design was fixed without peeking at those outcomes. To see exactly what is learned, assume $K \mid x, B \sim \mathrm{Binom}(B, \pi_j(x))$ with B determined before outcomes. For a fixed x, B, the expected unregularised loss minus its value at $p = \pi_j(x)$ is

$$\frac{B}{B_\bullet} D_{\mathrm{KL}}\left(\mathrm{Bern}(\pi_j(x)) \,\|\, \mathrm{Bern}(p)\right) \geq 0. \tag{57}$$

Thus the ideal unrestricted conditional predictor is the attack success frequency $\pi_j(x)$ on the *development* distribution; finite data, regularisation, optimiser error and population shift prevent equality in an actual trained model. No neural output is a confidence limit on an individual or a new population.

The gradient is explicit. With $P = (p_{ij})$, $K = (K_{ij})$ and $B = (B_{ij})$, the output-logit derivative is $\Delta^L = (B \odot P - K)/B_\bullet$. For the hidden layers, writing $Z^\ell = H^\ell W_\ell + \mathbf{1} b_\ell^\top$, reverse differentiation gives

$$\nabla_{W_{\mathrm{o}}}\mathcal{L} = (H^L)^\top \Delta^L + \lambda W_{\mathrm{o}}, \qquad \nabla_{b_{\mathrm{o}}}\mathcal{L} = \mathbf{1}^\top \Delta^L, \tag{58}$$

$$\Delta^\ell = (\Delta^{\ell+1} W_{\ell+1}^\top) \odot \mathbf{1}_{Z^\ell > 0}, \qquad \nabla_{W_\ell}\mathcal{L} = (H^\ell)^\top \Delta^\ell + \lambda W_\ell. \tag{59}$$

The second line uses $W_L = W_{\mathrm{o}}$ at its boundary; more formally the backward recursion starts from the output delta, propagates through each matrix and multiplies by the corresponding ReLU derivative. Appendix E gives a fully indexed version and the dimension check. The package includes a finite-difference check of deep-network derivatives.

15.3 Training, scale and falsification

The network uses seeded mini-batches, Adam, gradient clipping and early stopping on a separate held-out programme. Depth zero is trained on the same counted outcomes as a logistic baseline. For N development rows and E epochs, the dense leading arithmetic order is $O\{EN[dq + (L - 1)q^2 + qn]\}$ for $L \geq 1$, with $O\{dq + (L - 1)q^2 + qn\}$ weights; the logistic case has $O(ENdn)$ arithmetic. This is a computational scaling statement, not a statistical sample-size promise. The final Cn confidence bounds and independent attack trials remain necessary whatever the prediction speed. Unknown groups and out-of-domain features trigger a new assessment rather than an extrapolated certificate.

A valid empirical evaluation must compare deep and logistic rankings at fixed shortlist sizes and audit budgets on held-out *real* programmes; report log loss, Brier score, worst-group calibration, selected utility and false gates; repeat pooled-vintage attacks; and show uncertainty across programmes. This package currently provides a seeded synthetic end-to-end demonstration, not such an evaluation. Table 14 is regenerated by `scripts/sentry_r_benchmark.py` from 320 training rows and 100 rows in a separate toy programme; both models use the same counted outcomes. Its illustrative output must not be read as evidence of better release decisions.

Table 14: Seeded synthetic SENTRY-R illustration (320 development rows and 100 held-out rows); smaller validation loss is better. These are not results on real release programmes.

Proposal model	Log loss	Brier score	Best epoch
Logistic	0.3436	0.0970	80
Three-layer ReLU	0.2780	0.0817	63

16 Falsification and validation

Reproducing the equations of Sections 2 to 9 in software is an implementation check, not empirical validation. Evidence for the scenario model must be capable of contradiction. We propose six tests, each with a stated failure consequence, and report how the simulation of Section 10 performs against them.

1. **Linkage calibration.** Posterior probabilities from (10) must agree with held-out match frequencies within prespecified bands. Figure 3 shows the test passing for the correct model with a mean gap of 0.007 and failing for a misspecified covariate precision with a gap of 0.053, while the two risk estimates differ by 0.001. Failure means the risk figure is not supported, whatever its value.

2. **Population-frequency validation.** The model behind (16) must predict held-out or externally benchmarked key frequencies. In Section 10 the fitted law recovers the mean population cell size to within 1 per cent and the journalist risk to between 13 and 18 per cent, and the direction of the error is conservative. Failure means the journalist figure should be replaced by a bound.

3. **Adversary stress test.** A materially stronger plausible feature set must not push the

upper risk bound above τ. In the simulation, an adversary whose covariate is twice as precise raises the simulated risk by between 1 and 76 per cent across the 45 settings of the grid, with a median of 40 per cent. This is why the robust risk (25), and not the nominal risk, is the coordinate in Figure 5. Capability and modelling are separate things, and a second check records the difference: an adversary that keeps its information but assumes the covariate is noisier than it is loses only 0.007 of its success rate while its calibration gap grows sevenfold, so a risk figure alone will not reveal the mistake. The record swapping lever described in Section 10 was removed on the same principle. Failure of the test means the release fails or the scenario is revised in writing.

4. **Mechanism monotonicity.** Where stronger protection is claimed to reduce risk, violations must be investigated rather than smoothed. Every row of Table 4 violates the nominal bound on the supremum control, and the investigation of that violation is Proposition 2.

5. **Composition test.** Simulated joint attacks under (37) must be compared with the marginal bound (36) and evaluated against a programme tolerance. In Section 10.6 the comparison fails by a factor of 1.6 at two releases. Failure means the programme, not the release, is the object that needs governing.

6. **Temporal transport.** An assessment must fail or be re-estimated when new registers, linkage methods or data vintages move observed attack performance outside its prespecified range. This is the test that the expiry date d in Definition 2 enforces, and it is the only one of the six that cannot be run at assessment time.

17 Operational protocol

The governance record should follow five stages, in this order, with the output of each written down before the next begins.

1. **Declare** the attack scenario a, the target set T, the harm class h, the uncertainty set H_a and the loss $\mathcal{L}$.

2. **Compute** the point estimate $\widehat{R}_a$, the bound $\overline{R}_{1-\alpha}$ and the number of replicates B, with the dependence structure of the replicates stated.

3. **Review** whether the declared scenario is plausibly conservative, using the tests of Section 16.

4. **Apply** the tolerance τ through the named owner o, using rule (29).

5. **File** the complete record with the expiry date d and evidence sufficient to reproduce the computation.

Exposure and harm remain distinct throughout. The risk R_a measures the probability of successful exposure; the tolerance $\tau(h)$ reflects the consequence class, the legal duties and the

public value of the release. The owner owns the residual risk. The analyst owns the correctness of the calculation. These are separate responsibilities and should sit with separate people.

For a programme, two further items belong in the record: the joint attack simulation under (37) with its own tolerance, and, where a formal mechanism is used, the accountant's running total with the accounting method named. Neither can be inferred from the single-release records.

18 Discussion

The argument of this paper is that a release decision has three parts that should be separately visible, and that most of the difficulty in practice comes from collapsing them. Section 3 shows that the standard risk measures are theorems about one linkage experiment, so that changing the experiment changes the measure; Section 4 shows that the resulting figure is an estimate whose uncertainty belongs in the decision rule; Section 5 shows that the tolerance is a policy quantity that a stated price of harm already pins down. The two negative results are the ones with the most immediate consequence for institutions. A k-anonymous release does not guarantee a risk of $1/k$ for the people in it once the adversary holds anything beyond the key (Table 4), and a programme of low-risk releases can be attacked far above the bound its marginals imply (Table 6). Section 13 shows what the second costs on a release system rather than in a construction: the policies a supervisor operates are calibrated on one release, and every one of them leaves the pooled programme above the tolerance it was written for.

Four limitations should be stated plainly. First, the simulation is synthetic. It establishes that the phenomena occur under a transparent data generating process and that the estimators behave as the propositions say, and it cannot establish the magnitudes an agency would see on its own register; those depend on the auxiliary data actually available, which is exactly the quantity a declared scenario has to guess. Second, the framework prices harm through a scalar c_h within a declared class h. Harms that fall on groups rather than individuals, and harms that arise from correct inference about a population rather than identification of a person, are outside it, and the second of these is precisely what Dwork and Roth [2014] argue is not a privacy violation at all. Third, the joint attack in Section 10.6 is one pooling rule among many. It gives a lower bound on what a pooling adversary achieves, so it supports the negative result of Proposition 10, but no simulation can deliver the supremum over rules that a formal guarantee delivers by construction. Fourth, the neural proposal model is illustrated only on independent synthetic attack trials. Its output has no established calibration or utility benefit on the credit and census data reported here, and the final audit remains conditional on the declared attack scenario.

Four extensions look tractable. The frontier estimation in Section 6 could be given a proper inferential treatment, since the frontier is a functional of an estimated set and its selected point is biased in the same way as any quantity chosen by maximising an estimate. The risk budget of Proposition 11 could be extended from the union event to a joint accountant that admits dependence, which would need a composition theorem for scenario-conditional risk under a stated dependence class. The implied tolerance of Proposition 6 could be estimated from revealed institutional behaviour, since agencies have been making these decisions for decades

and their past choices identify κ under the framework. And the calibration test of Section 16 could be developed into a formal specification test with a distribution under the null.

The recommendation is unchanged from the short version of this argument. Publish, alongside each release decision, the risk estimate, the uncertainty bound and the number of replicates behind it, the tolerance, the adversary specification, the mechanism setting, the named owner and the expiry date. For a programme, publish the joint simulation as well. That record does not remove judgement from the decision. It puts judgement in the policy quantity where it belongs and leaves the probability calculation reproducible and open to being proved wrong.

18.1 What a release committee should take from this

Three decisions follow from the argument, and they are the only three that need a committee. The first is the tolerance, its accountable owner and its expiry date, which Definition 2 makes a record rather than a habit. The second is whether that tolerance is asserted of each release or of the programme, which Section 13 shows is not a matter of wording: at a tolerance of 0.12 the per-release reading of the same rule permits a release the programme reading forbids, and the difference between them is 32,581 released relationships against 20,389. The third is the adversary specification, written down, so that the assumption can be disagreed with explicitly instead of inherited silently.

A fourth choice is easy to miss because the arithmetic hides it. The measures of Section 3.3 protect different people. The marketer measure is an average over the released records, and a release can meet a tolerance on that average while a particular record in it is identified with near certainty. The prosecutor measure in its supremum form protects the most exposed record instead, and the subgroup bounds of Section 13 are the operational form of the same idea. Which one an institution reports is an ethical choice about whose protection the threshold is for, not a technical preference, and the framework supports either as long as the committee says which it has chosen.

Data and code availability

The numerical results and figures in the original sections are generated by this repository's experiment pipeline, with the configuration and recorded outputs in `artifacts-paper/`. The manuscript's numerical checks can be run with `scripts/check_paper_numbers.py`; numerical reproducibility is a separate claim from the new neural method. The new `privacy_risk.neural_trust` module implements a vector risk prediction, a finite-shortlist proposal and an independent simultaneous binomial audit. Its seeded demonstration is synthetic and produces no inputs to the original empirical tables or figures. Validation of neural ranking on held-out real release programmes remains future work. The two public microdata files described in Section 11 are loaded through the `wooldridge` package by the original pipeline.

Acknowledgements

An earlier version was prepared for the Workshop on Data Security and Privacy of the Banco de Portugal and the Irving Fisher Committee, Lisbon, September 2026. The views are the author's

own.

A Proofs

Proof of Proposition 1. Condition on $S_t = 1$ and $K_a(X_t) = k$. By Assumption 1 the released key of every record equals its confidential key, and by Assumption 2 the adversary observes k. The compatible set is therefore $\{j \in I : K_a(X_j) = k\}$, which has cardinality f_k by (3) and contains t. By Assumption 3 the adversary's information is invariant to permutations of that set, and by Assumption 4 it selects each member with probability $1/f_k$. Exactly one member is correct, so

$$\mathbb{P}\{\delta_a = t \mid S_t = 1, K_a(X_t) = k, f_k\} = \sum_{j=1}^{f_k} \frac{1}{f_k} \mathbf{1}\{j = t\} = \frac{1}{f_k}. \qquad \square$$

Proof of Proposition 2. By (12) and Proposition 1, $\sup_{t \in I} R_{\text{pros}}(t, \theta) = 1/\min_{k: f_k > 0} f_k$. Hence $\sup_t R_{\text{pros}} \leq \tau$ if and only if $\min_k f_k \geq 1/\tau$, and since f_k is an integer this is $\min_k f_k \geq \lceil 1/\tau \rceil$, which is the definition of k-anonymity with $k = \lceil 1/\tau \rceil$.

For the failure claim, drop Assumption 3. Let the release publish a confidential variable y alongside the key, let a key class have $f_k = k$ members with distinct values of y, and let the adversary observe $z_t = y_t + \sigma\xi$ with ξ standard normal and σ small relative to the within-class spacing of y. The posterior (10) concentrates on the member whose y is nearest z_t, and as $\sigma \downarrow 0$ the success probability of the maximum-posterior rule tends to one, while k-anonymity holds for every σ. Therefore $\sup_t R_a^{\text{in}} > 1/k$ for all sufficiently small σ. $\square$

Proof of Proposition 3. With zero successes the one-sided Clopper–Pearson upper limit solves $(1 - U)^B = \alpha$, giving $\overline{R}_{1-\alpha} = 1 - \alpha^{1/B}$ as in (21). Then $\overline{R}_{1-\alpha} \leq \tau$ if and only if $\alpha^{1/B} \geq 1 - \tau$. Taking logarithms, and noting $\log \alpha < 0$ and $\log(1 - \tau) < 0$, this is $B^{-1} \log \alpha \geq \log(1 - \tau)$, that is $B \geq \log \alpha / \log(1 - \tau)$. Since $\log(1 - \tau) = -\tau + O(\tau^2)$, the right side is $-\log \alpha/\tau + O(1)$ as $\tau \downarrow 0$. $\square$

Proof of Proposition 4. For any occupied key class, $f_k \leq F_k$ by (3), hence $1/f_k \geq 1/F_k$ pointwise, and monotonicity of expectation gives the first claim, with equality if and only if $f_k = F_k$ almost surely for every class the average touches. The second claim is Jensen's inequality applied to the convex function $x \mapsto 1/x$ on $(0, \infty)$, which is strict unless f_k is degenerate. $\square$

Proof of Proposition 5. At $R_a = \tau$, asymptotic symmetry of the estimator gives $\mathbb{P}(\widehat{R}_a \leq R_a) \to 1/2$, so the point-estimate rule releases with probability tending to $1/2$. A valid one-sided $(1-\alpha)$ upper bound satisfies $\mathbb{P}\{R_a \leq \overline{R}_{1-\alpha}\} \geq 1 - \alpha$, hence $\mathbb{P}\{\overline{R}_{1-\alpha} < R_a\} \leq \alpha$. At the boundary, release under (29) implies $\overline{R}_{1-\alpha} \leq \tau = R_a$, an event of probability at most α up to equality conventions. $\square$

Proof of Proposition 6. On the frontier the problem is $\max_r \{\lambda\varphi(r) - c_h r\}$ over $[\underline{r}, \overline{r}]$. The objective is strictly concave because φ is, so a maximiser exists and is unique. Its derivative is $\lambda\varphi'(r) - c_h$, which is strictly decreasing; under $\varphi'(\underline{r}) > \kappa > \varphi'(\overline{r})$ with $\kappa = c_h/\lambda$ it is positive at $\underline{r}$ and negative at $\overline{r}$, so the maximiser is interior and solves $\varphi'(r) = \kappa$. Monotonicity of $\tau^\star$ in κ follows because φ' is strictly decreasing, so $\tau^\star = (\varphi')^{-1}(\kappa)$ is decreasing in κ. $\square$

Proof of Proposition 8. The feasible set $\Theta_\tau = \{\theta : \sup_t \overline{R}_{1-\alpha}(t,\theta) \le \tau\}$ is closed by lower semicontinuity of the constraint function, hence compact as a closed subset of the compact Θ, and nonempty by assumption. An upper semicontinuous function attains its maximum on a nonempty compact set. □

Proof of Proposition 9. Monotonicity gives $C_i \subseteq \bigcup_j C_j$ and hence $R_i \le R_\cup$ for every i, so $\max_i R_i \le R_\cup$. Boole's inequality gives $R_\cup \le \sum_i R_i$, and $R_\cup \le 1$ always, which is the upper bound. Attainment: if the events are nested, $C_1 \supseteq C_2 \supseteq \cdots$, then $R_\cup = R_1 = \max_i R_i$. If they are disjoint and $\sum_i R_i \le 1$, then $R_\cup = \sum_i R_i$; if $\sum_i R_i > 1$, a standard overlapping construction on $[0,1]$ with Lebesgue measure, placing C_i as intervals that cover $[0,1]$, gives $R_\cup = 1$. □

Proof of Proposition 10. Fix $m \ge 2$ and an integer $k \ge 1/\nu$. Let the sample consist of $n = k^m$ units labelled by the vectors $x = (x_1, \ldots, x_m) \in \{1, \ldots, k\}^m$, all present with certainty. Let release i publish the key $K_i(x) = (x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_m)$, that is all coordinates except the ith. Each key class of release i contains exactly k units, so the release is k-anonymous and, under Assumptions 1 to 4, $R_i = 1/k \le \nu$.

Now let the adversary hold all m releases and know the target's full label. The candidate set from release i is $\{x : K_i(x) = K_i(t)\}$, which is the set of units agreeing with t in all coordinates except possibly the ith. The intersection over $i = 1, \ldots, m$ of these sets is $\{t\}$, because a unit in the intersection agrees with t in every coordinate: for coordinate j, membership in the candidate set of any release $i \ne j$ forces $x_j = t_j$, and $m \ge 2$ guarantees such an i exists. Hence $\delta_a^{(m)}$ that selects the unique element of the intersection satisfies $R_{\text{pool}} = 1$. Since ν was arbitrary and $(R_1, \ldots, R_m) = (1/k, \ldots, 1/k)$ can be made arbitrarily small while $R_{\text{pool}} = 1$, no function of the marginals can bound R_{pool} from above.

For the final claim, let $A_i(t)$ be the candidate set of release i and $G_m(t) = \bigcap_{i \le m} A_i(t)$, and suppose the adversary selects uniformly from $G_m(t)$. Then $G_{m+1}(t) \subseteq G_m(t)$ and $t \in G_m(t)$ for every m, so $1/|G_{m+1}(t)| \ge 1/|G_m(t)|$ pathwise and taking expectations gives monotonicity. Since $G_m(t) \subseteq A_i(t)$ for each $i \le m$, the same argument gives $R_{\text{pool}} \ge \max_{i \le m} \mathbb{E}[1/|A_i(t)|] = \max_i R_i$. □

Proof of Proposition 11. The objective is strictly concave and the constraint set is compact and convex, so a unique maximiser exists. Since φ is increasing, the constraint binds. The Lagrangian is $\sum_i w_i \varphi(r_i) - \mu(\sum_i r_i - \tau_{\text{prog}})$ and the first-order conditions are $w_i \varphi'(r_i) = \mu$ for every i, which is (38); concavity makes them sufficient. For $\varphi = \log$ the conditions read $w_i / r_i = \mu$, so $r_i \propto w_i$, and the constraint fixes the constant at $\tau_{\text{prog}} / \sum_j w_j$. For $\varphi(r) = a + b r^\gamma$ with $b < 0$ and $\gamma < 0$, which is increasing and strictly concave on $(0, \infty)$, we have $\varphi'(r) = b\gamma r^{\gamma - 1}$, so (38) reads $w_i r_i^{\gamma - 1} = \mu / (b\gamma)$ and hence $r_i \propto w_i^{1/(1-\gamma)}$, which is (39). □

Proof of Proposition 12. At posterior probability p, acting has expected payoff $V(p) = ph - (1-p)c_f - c_a = p(h + c_f) - (c_f + c_a)$. Since $h + c_f > 0$, $V(p) \ge 0$ if and only if $p \ge (c_f + c_a)/(h + c_f)$. If this ratio exceeds one, no posterior below certainty justifies the attack, which gives the truncation in (41). □

Proof of Proposition 13. For fixed $u \in (0,1)$, the Bernoulli likelihood ratio through time B between risk u and risk r_0 is a nonnegative r_0-martingale with initial value one. Integrating these

likelihood-ratio martingales against the Beta(a, b) density preserves the martingale property by Tonelli's theorem and gives exactly (42). Ville's inequality for a nonnegative martingale yields (43). The event that r_0 exits $\mathcal{C}_B$ at some time is the event that $E_B(r_0) \geq 1/\alpha$ at some time, so its probability is at most α. □

Proof of Proposition 14. Let x and x' be the databases in which H is true and false respectively; they differ in one individual's record, so they are neighbouring, and by hypothesis the remaining records have the same distribution under both. For any observed output set E of positive probability, (46) with $\delta_{\mathrm{dp}} = 0$ gives the likelihood ratio bound $\mathbb{P}\{M(x) \in E\}/\mathbb{P}\{M(x') \in E\} \leq e^{\varepsilon}$. Bayes' rule in odds form gives

$$\frac{\mathbb{P}(H \mid M \in E)}{1 - \mathbb{P}(H \mid M \in E)} = \frac{p}{1-p} \cdot \frac{\mathbb{P}\{M(x) \in E\}}{\mathbb{P}\{M(x') \in E\}} \leq \frac{p}{1-p} e^{\varepsilon},$$

and solving for $\mathbb{P}(H \mid M \in E)$ gives (47). The bound extends to $\delta_{\mathrm{dp}} > 0$ with an additive term of order $\delta_{\mathrm{dp}}/\mathbb{P}\{M(x') \in E\}$, which is why δ_{dp} is usually set below the reciprocal of the sample size. □

B Expanded derivations

This appendix expands the calculations used in the text, so that each reported quantity can be reconstructed without treating an intermediate identity as given.

B.1 From joint success to membership-conditional risk

Let $A_t = \{\delta_a(Z_t, Y_\theta) = t\}$ and $B_t = \{S_t = 1\}$. Since a released record cannot equal an unsampled target, $A_t \subseteq B_t$, and

$$R_a^{\mathrm{unc}}(t, \theta) = \mathbb{P}(A_t \cap B_t \mid \mathcal{B}_a) = \mathbb{P}(A_t \mid B_t, \mathcal{B}_a)\mathbb{P}(B_t \mid \mathcal{B}_a) = \pi_t R_a^{\mathrm{in}}(t, \theta), \tag{60}$$

which is (8). If the auxiliary information changes the adversary's belief about membership, the last factor is $\mathbb{P}(B_t \mid \mathcal{B}_a)$ and not the design probability π_t; using π_t requires that $\mathcal{B}_a$ carry no information about who was sampled.

B.2 Bayesian linkage from comparison vectors

Let $\gamma_{tj} = \gamma(Z_t, Y_j)$ be a comparison vector, and write $m(\gamma) = p(\gamma \mid H_{tj} = 1)$ and $u(\gamma) = p(\gamma \mid H_{tj} = 0)$ for its density under a match and a non-match, with likelihood ratio $\Lambda_{tj} = m(\gamma_{tj})/u(\gamma_{tj})$. Under the usual assumptions that the fields of a comparison vector are conditionally independent given the match status [Fellegi and Sunter, 1969], and that the comparison vectors of distinct candidates are conditionally independent of one another, Bayes' theorem and division by $\prod_\ell u(\gamma_{t\ell})$ give

$$p_{tj} = \frac{q_{tj}\Lambda_{tj}}{q_{t0}\Lambda_{t0} + \sum_{\ell \in I} q_{t\ell}\Lambda_{t\ell}}, \tag{61}$$

where $\Lambda_{t0} = 1$, because under the no-match hypothesis every candidate is scored by its non-match density and the division leaves one. This is (10) with $L = \Lambda$. Exact exchangeable

matching is the case of equal likelihood and prior weight on each of the f_k compatible records and zero weight elsewhere, giving $1/f_k$ for each. For two candidates,

$$\frac{p_{tj}}{p_{t\ell}} = \frac{q_{tj}}{q_{t\ell}} \cdot \frac{L_{tj}}{L_{t\ell}}, \tag{62}$$

so the reciprocal-frequency figure is valid only when both ratios equal one inside the key class. Equation (62) is where informative side information enters.

B.3 Journalist risk under binomial thinning

Conditional on $F_k = v$, Bernoulli sampling gives $f_k \mid F_k = v \sim \text{Binomial}(v, \pi)$, so $\mathbb{P}(F_k = 1, f_k = 1) = \pi\mathbb{P}(F_k = 1)$ and $\mathbb{P}(f_k = 1) = \sum_{v\geq 1} v\pi(1-\pi)^{v-1}\mathbb{P}(F_k = v)$. Dividing the first by the second gives (16). For a parametric frequency law $g(v;\eta)$ the plug-in estimator is

$$\widehat{\rho}_k = \frac{\pi g(1;\widehat{\eta})}{\sum_{v\geq 1} v\pi(1-\pi)^{v-1} g(v;\widehat{\eta})}, \tag{63}$$

and the analyst must carry the uncertainty in $\widehat{\eta}$ through (27), or refit g inside each replicate.

B.4 Exact binomial bounds and simultaneous control

If $X = \sum_b C_t^{(b)} = x$, the one-sided Clopper–Pearson upper endpoint is

$$\overline{R}_{1-\alpha}(x, B) = \text{Beta}^{-1}_{1-\alpha}(x+1, B-x), \qquad x < B, \tag{64}$$

and one when $x = B$, where $\text{Beta}^{-1}_q(a,b)$ is the qth quantile of a beta distribution. At $x = 0$ the beta distribution function is $F(u) = 1-(1-u)^B$, so $F(\overline{R}) = 1-\alpha$ recovers (21).

If T holds q targets and the decision uses their maximum, separately valid $(1-\alpha)$ intervals do not give a $(1-\alpha)$ file-level statement. Bonferroni control uses targetwise error α/q. Section 10.4 needs the lower form, since it reports the worst target, so writing $\underline{R}_{\alpha/q,t}$ for a one-sided lower limit,

$$\mathbb{P}\left\{R_t \geq \underline{R}_{\alpha/q,t} \text{ for every } t \in T\right\} \geq 1 - \sum_{t\in T} \frac{\alpha}{q} = 1-\alpha, \tag{65}$$

and the same argument gives the upper form when the decision uses a maximum. This is conservative but valid without independence, and it is what Section 10.4 uses for the worst target. A max-statistic bootstrap can replace it when the resampling scheme preserves dependence across targets [Efron and Tibshirani, 1993].

B.5 Nested estimator for simulation and input uncertainty

Draw outer replicates $s = 1, \ldots, S$. In each, redraw the confidential sample according to its design, refit any auxiliary model, apply the mechanism, and run B_s conditional attacks. With $\widehat{R}_s = B_s^{-1}\sum_b C_{sb}$ and $\overline{R} = S^{-1}\sum_s \widehat{R}_s$, an analysis-of-variance decomposition is

$$\widehat{V}_{\text{within}} = \frac{1}{S}\sum_{s=1}^{S} \frac{\widehat{R}_s(1-\widehat{R}_s)}{B_s}, \qquad \widehat{V}_{\text{between}} = \frac{1}{S-1}\sum_{s=1}^{S}(\widehat{R}_s - \overline{R})^2 - \widehat{V}_{\text{within}}, \tag{66}$$

the second truncated at zero when the finite estimate falls slightly negative. The variance of $\overline{R}$ is then approximately $(\widehat{V}_{\text{between}} + \widehat{V}_{\text{within}})/S$ for independent outer replicates. Percentile or studentised outer bootstrap bounds avoid relying on normality near zero. Section 10.9 reports both terms.

B.6 Tangency and boundary cases

On a differentiable frontier $u = \varphi(r)$, maximise $J(r) = \lambda\varphi(r) - c_h r$. Then

$$J'(r) = \lambda\varphi'(r) - c_h, \qquad J''(r) = \lambda\varphi''(r) < 0, \tag{67}$$

so the first-order condition is sufficient and gives $\varphi'(r) = c_h/\lambda$. If $c_h/\lambda \geq \varphi'(\underline{r})$ the optimum is the safest boundary $\underline{r}$; if $c_h/\lambda \leq \varphi'(\overline{r})$ it is the highest-risk boundary $\overline{r}$. These cases complete Proposition 6 when its interiority condition fails.

For the programme allocation, allow $r_i = 0$ and attach multipliers ξ_i to the non-negativity constraints. The Karush–Kuhn–Tucker conditions are

$$w_i\varphi_i'(r_i) - \mu + \xi_i = 0, \quad \xi_i r_i = 0, \quad \mu\Big(\tau_{\text{prog}} - \sum_i r_i\Big) = 0, \quad \mu, \xi_i \geq 0, \tag{68}$$

so active releases equalise weighted marginal value, and a release receives zero budget only when its marginal value at zero is no greater than the shadow price [Boyd and Vandenberghe, 2004]. This is the general form behind (38).

B.7 Predictive value of an alleged relationship

Let D be the event that a stated relationship holds and A the event that the adversary alleges it. Bayes' theorem gives

$$\mathbb{P}(D \mid A) = \frac{ps}{ps + (1-p)(1-q)}, \qquad p = \mathbb{P}(D),\ s = \mathbb{P}(A \mid D),\ q = \mathbb{P}(A^c \mid D^c), \tag{69}$$

the sensitivity s and specificity q of the allegation. The quantity that matters for harm is the left side, not s. When p is small, as it is for borrower–lender pairs, a specificity close to one still leaves a modest predictive value, and Section 12.1 reports the four cells of the table so that (69) can be checked as an identity rather than believed.

B.8 Residual disclosure from published totals

Let cells c carry published totals $T_c = \sum_b x_{bc}$, where x_{bc} is contributor b's amount. Stacking the published totals gives a linear system $Ax = T$. A contribution x_{bc} is determined by the release if and only if it is constant on the feasible set $\{x : Ax = T,\ x \geq 0\}$ intersected with whatever else the adversary knows, which for a participating institution includes its own row. The non-negativity constraint is not decoration: it is what makes a zero total disclosive, and it is what linear-programming attacks on suppressed tables exploit. Two familiar cases follow at once. A cell with one contributor gives $x_{bc} = T_c$. A cell with two contributors gives the other's amount to either of them by subtraction. Requiring three contributors per published cell removes both,

which is why that rule is standard. Counting those two cases gives a lower bound on what the release determines, not the whole of it, and it does not address the differencing attack below.

B.9 Differencing across consecutive releases

Let $n_{c,m}$ and $T_{c,m}$ be the count and total published for cell c in month m. If $n_{c,m+1} - n_{c,m} = 1$ and exactly one relationship moved, the arriving amount is $T_{c,m+1} - T_{c,m}$. The adversary does not observe the number of moves. Under a model in which arrivals A and departures D are independent Poisson counts with mean λ, the posterior that an observed unit change came from a single move is

$$\mathbb{P}(A = 1, D = 0 \mid |A - D| = 1) = \frac{\mathbb{P}(A = 1)\mathbb{P}(D = 0)}{\sum_{|a-d|=1} \mathbb{P}(A = a)\mathbb{P}(D = d)}. \tag{70}$$

The inference is therefore probabilistic, and its strength falls as the cell becomes busier. Section 12.1 evaluates (70) on the simulated register.

C Simulation details

Generation. District and occupation categories are drawn from distributions with probabilities proportional to $j^{-0.9}$ and $j^{-0.8}$ respectively, which produces the skewed cell sizes and the population uniques reported in Section 10. Ages are drawn uniformly on $[18, 88)$ and then thinned towards younger ages, so that the oldest bands are the sparse ones. All draws use a single seeded generator.

Attack. Targets are drawn uniformly with replacement from the released records, except in the per-target experiment of Section 10.4, where each named target is attacked 250 times. The reported lower bound for the worst target is the one-sided Clopper–Pearson limit from those 250 replicates, taken at 0.05 divided by the number of targets examined, because that target is selected on its own estimate. The adversary blocks on the target's district and sex, which it observes without error. Within the block it evaluates the likelihood of the observed age band and occupation group under the error model that generated them, with the neighbour count of each candidate band handled so that the declared model and the generating model agree at the boundary bands, and multiplies by the normal density of the noisy covariate. It selects the maximum-posterior candidate and abstains only when the total likelihood vanishes. Replicates within a setting are independent draws of the target and of the adversary's information, so (20) and the Clopper–Pearson bound apply to that draw, conditional on the realised sample and mechanism output. They do not cover sampling variability, and an institution reusing one auxiliary file across replicates would not have even this much independence.

Grid and frontier. The 45 mechanism settings are evaluated with 1,200 replicates each for both members of the uncertainty set, and the robust risk is the larger point estimate with its own bound. Utility is the negative of the average of three root mean squared errors, for the mean of y by district, by five-year age band and by occupation, each computed against the

population truth over target categories that the release can report with at least three records and normalised by the standard deviation of y, plus a penalty of 0.5 times the average share of target categories left unreportable. Age and occupation estimates are formed from the released classification, so a release with 20-year bands or three occupation groups can only supply a piecewise constant answer to an analysis stated at five-year bands or twelve occupations. The frontier is the nondominated subset in the plane of the risk bound and this utility. In the allocation of Section 10.6 the fitted curve is evaluated on $(0, \tau_{\text{prog}}]$ with a numerical lower bound of 10^{-3} on each share, which is inactive at the reported optimum.

Credit register. The register of Section 12.1 draws borrowers with a sector, region and size class, gives each between one and four lending relationships with larger firms holding more, and draws exposures lognormally within size class and arrears at rates that vary by sector and size. Six monthly vintages are generated by giving each relationship a start month and a duration. The linkage attack blocks on region, requires an exact match on the sector and size class, and multiplies by the density of a reading of log exposure with standard deviation 0.6. Success is scored at the relationship, so recovering a different relationship of the correct borrower counts as a failure; the borrower-level figure is reported beside it. The published aggregates are totals by region, sector and size class, and the differencing comparison uses the counts of those cells across consecutive vintages, which is a separate lever from the minimum cell count applied to the loan-level file.

Composition. The four releases share one sample, so all dependence between them is the dependence an agency would face when publishing several products from one collection. Candidate sets are the key classes of each release after suppression. A target suppressed from a release contributes zero to that release's marginal risk and does not narrow the intersection, so the pooled figure is what an adversary achieves using the releases that contain the target. Reported marginal and pooled risks are means of $1/|A_i(t)|$ and $1/|G_m(t)|$ over the same 20,000 targets rather than Bernoulli draws, which removes selection noise without changing the estimand.

D Mathematical details for the neural trust position

Experiment and estimand. Let $\mathcal{D}$ contain all development observations, candidate generation, neural fitting, tuning, scenario definitions and proposed releases. Conditional on $\mathcal{D}$, the shortlist Θ_0, its cardinality C, and all sample sizes $B_{\theta j} \geq 1$ are fixed. An audit draw $Z_{\theta j i}$ includes the target, auxiliary data when freshly sampled, and any randomness in the declared attack. The measurable success indicator $S_{\theta j i} = s_{\theta j}(Z_{\theta j i})$ is one exactly when the declared harm criterion is met. For fixed (θ, j) assume the draws are identically distributed and independent conditional on $\mathcal{D}$, with law $P_{\theta j}$. Thus

$$r_j(\theta) = \int s_{\theta j}(z)\, dP_{\theta j}(z), \qquad K_{\theta j} = \sum_{i=1}^{B_{\theta j}} S_{\theta j i} \mid \mathcal{D} \sim \text{Binom}(B_{\theta j}, r_j(\theta)). \tag{71}$$

The risk is conditional on a frozen release and on the audit design. If target records or auxiliary files are reused, the resulting dependent observations must be treated as one sampling unit or justified by another valid confidence procedure. Independence across distinct (θ, j) pairs is unnecessary. For a finite population sampled without replacement, the binomial assumption is inappropriate unless the attack design supplies independent replacement draws; a hypergeometric or design-based bound is needed instead.

Exact inversion. Write $b = B_{\theta j}$, $k = K_{\theta j}$, $\gamma = \alpha/(Cn)$, and $F_p(k) = \sum_{t=0}^{k} \binom{b}{t} p^t (1-p)^{b-t}$. For $k < b$, $F_p(k)$ is continuous and strictly decreasing in $p \in (0,1)$. The upper endpoint $u(k)$ is the unique solution of $F_{u(k)}(k) = \gamma$; set $u(b) = 1$. The binomial–beta identity gives

$$F_p(k) = 1 - I_p(k+1, b-k), \qquad u(k) = F^{-1}_{\mathrm{Beta}(k+1,b-k)}(1-\gamma). \tag{72}$$

For any fixed true p, the event $\{u(K) < p\}$ is a lower-tail set of integer counts: by monotonicity it equals $\{F_p(K) < \gamma\}$, apart from the harmless equality boundary. If k_* is the largest count in this set, $\Pr_p\{u(K) < p\} = F_p(k_*) < \gamma$; if the set is empty the probability is zero. Hence $\Pr_p\{p \le u(K)\} \ge 1-\gamma$ for every $p \in [0,1]$. The possible strict inequality reflects the discreteness of the count.

Familywise selection. Define the event

$$E = \bigcap_{\theta \in \Theta_0} \bigcap_{j \in \mathcal{J}} \{r_j(\theta) \le \overline{r}_{\theta j}\}. \tag{73}$$

Conditional on every development history for which the assumptions hold, $\Pr(E^c \mid \mathcal{D}) \le \sum_{\theta,j} \gamma = \alpha$. Any measurable decision rule $\widehat{\theta}$ using the complete vector of final bounds, including the choice of the greatest utility among accepted candidates, satisfies the deterministic implication

$$E \implies \left[\widehat{\theta} \text{ accepted}\right] \implies \max_{j \in \mathcal{J}} \frac{r_j(\widehat{\theta})}{\tau_j} \le 1. \tag{74}$$

The probability of a false certification is therefore at most α conditionally and, by $\Pr(E^c) = \mathbb{E}[\Pr(E^c \mid \mathcal{D})]$, unconditionally. This is a frequentist guarantee for the declared finite family; it is not a posterior probability that a particular accepted release is safe. An audit outcome used to add candidates, alter scenarios, retrain the network or select sample sizes destroys the conditioning argument unless a separate sequential or adaptive design is proved valid.

Budget and an illustrative calculation. At $k = 0$, equation (72) simplifies to $u(0) = 1 - \gamma^{1/b}$. Consequently $u(0) \le \tau$ if and only if $(1-\tau)^b \le \gamma$, equivalently

$$b \ge \left\lceil \frac{\log \gamma}{\log(1-\tau)} \right\rceil. \tag{75}$$

For illustration only, $C = 4$, $n = 2$, $\alpha = 0.05$ and $\tau = 0.12$ give $\gamma = 0.00625$ and $b = 40$. The zero-success endpoint is approximately 0.11916 at $b = 40$, but approximately 0.12202 at $b = 39$. These numbers do not establish the risk of any real release. The total number of attack trials is

$\sum_{\theta,j} B_{\theta j}$; when each of Cn pairs uses the same b, it is Cnb. For small τ, $-\log(1-\tau) = \tau + O(\tau^2)$, so the zero-success requirement grows as $\log(Cn/\alpha)/\tau$ per pair. The neural output dimension changes this multiplicity even when its input feature dimension d stays fixed.

Neural objective and its limits. Let $\pi_j(x) = \Pr(Y_j = 1 \mid X = x)$ in the development distribution. For a positive weight $w_j(x)$ depending only on x and j, the conditional expected binary cross entropy at a predicted probability q is

$$\ell_x(q) = -w_j(x)[\pi_j(x)\log q + (1-\pi_j(x))\log(1-q)], \quad \ell_x'(q) = w_j(x)\frac{q-\pi_j(x)}{q(1-q)}, \tag{76}$$

with $\ell_x''(q) > 0$ at its unique minimum $q = \pi_j(x)$ whenever $0 < \pi_j(x) < 1$. Outcome-dependent weights instead change that minimiser. For the implemented one-hidden-layer ReLU model, with row-vector features $X \in \mathbb{R}^{N\times d}$, width q, and n outputs, put $H = (XW_1 + \mathbf{1}b_1^\top)_+$ and $\widehat{R} = \sigma(HW_2 + \mathbf{1}b_2^\top)$. For the objective $L = -N^{-1}\sum_{i,j} w_{ij}[y_{ij}\log\widehat{r}_{ij} + (1-y_{ij})\log(1-\widehat{r}_{ij})]$, the output-logit derivative is $G = (W \odot (\widehat{R} - Y))/N$. At differentiability points of ReLU,

$$\nabla_{W_2} L = H^\top G, \quad \nabla_{b_2} L = \mathbf{1}^\top G, \quad \nabla_{W_1} L = X^\top[(GW_2^\top) \odot \mathbf{1}_{XW_1 + \mathbf{1}b_1^\top > 0}]. \tag{77}$$

These identities explain the code's backpropagation; regularisation or optimisation does not make its predictions exact. Under the fitted weights, ReLU's Lipschitz constant of one and sigmoid's scalar derivative at most $1/4$ give the predictor sensitivity bound $\|\widehat{\boldsymbol{r}}(x) - \widehat{\boldsymbol{r}}(x')\|_2 \le \|W_2\|_2\|W_1\|_2\|x - x'\|_2/4$. It controls changes in *predictions*, not errors relative to actual risks, and is no substitute for the audit.

The input dimension. Suppose, as an additional assumption, the true conditional risk is L-Lipschitz in $\|\cdot\|_\infty$ on $[0,1]^d$. A grid of spacing $h \le \epsilon/L$ lets a value at a nearby grid point approximate the true risk to within ϵ, requiring at most $(1 + \lceil L/\epsilon \rceil)^d$ grid points. Estimating those values adds sampling cost at each point; the grid count alone already grows exponentially in d. If no regularity links unsampled inputs to observed inputs, two risk functions can agree on every observed feature vector and disagree arbitrarily at an unobserved vector, so there is no distribution-free uniform interpolation guarantee. Structural assumptions may help prediction, but the finite-family audit guarantee above does not rely on them.

Distribution shift. Using $\mathrm{TV}(P,Q) = \sup_A |P(A) - Q(A)|$ on the complete audit-draw space, the layer-cake representation for any success function $0 \le s \le 1$ yields $|\mathbb{E}_Q s - \mathbb{E}_P s| \le \mathrm{TV}(P,Q)$. If a justified deterministic bound $\mathrm{TV}(P_{\theta j}, Q_{\theta j}) \le \delta_{\theta j}$ holds for all inspected pairs, a simultaneous deployed-risk bound is

$$\overline{r}_{\theta j}^{\,\mathrm{deploy}} = \min\{1, \overline{r}_{\theta j} + \delta_{\theta j}\}, \qquad T^{\mathrm{deploy}}(\theta) = \min_j \left(1 - \frac{\overline{r}_{\theta j}^{\,\mathrm{deploy}}}{\tau_j}\right). \tag{78}$$

The same familywise probability applies because the TV inequality holds pointwise on the simultaneous coverage event. A TV estimate, itself uncertain, needs its own valid upper bound

and error allocation. A distance between neural feature embeddings alone cannot establish a TV bound on the full experiment.

E SENTRY-R derivations and adaptive development

Corrected backpropagation indices. Let there be $L \geq 1$ hidden layers indexed $m = 1, \ldots, L$, $H^0 = X$, $Z^m = H^{m-1}W_m + \mathbf{1}b_m^\top$, $H^m = \rho(Z^m)$, and $A = H^L W_{L+1} + \mathbf{1}b_{L+1}^\top$. Here W_m has shape $q_{m-1} \times q_m$, with $q_0 = d$ and $q_{L+1} = n$. Let $D^{L+1} = (B \odot \sigma(A) - K)/B_\bullet$. The scalar derivative $\partial[-k \log \sigma(a) - (b-k)\log(1-\sigma(a))]/\partial a = b\sigma(a) - k$ follows by direct chain rule. Then, for $m = L, \ldots, 1$,

$$D^m = (D^{m+1}W_{m+1}^\top) \odot \mathbf{1}_{Z^m > 0}, \quad \nabla_{W_m}\mathcal{L} = (H^{m-1})^\top D^m + \lambda W_m, \quad \nabla_{b_m}\mathcal{L} = \mathbf{1}^\top D^m, \tag{79}$$

$$\nabla_{W_{L+1}}\mathcal{L} = (H^L)^\top D^{L+1} + \lambda W_{L+1}, \quad \nabla_{b_{L+1}}\mathcal{L} = \mathbf{1}^\top D^{L+1}. \tag{80}$$

D^{L+1} has shape $N \times n$; each D^m has shape $N \times q_m$, so every displayed matrix multiplication is dimensionally well defined. At a zero ReLU preactivation the stated convention takes the subgradient zero. The depth-zero baseline sets $H^0 = X$ and differentiates the single sigmoid output layer directly. Adam changes the update rule, not these gradients. This derivation exposes what the network computes without asserting that the learned weights identify a true attack probability.

Adaptive proposal rounds and a fixed final experiment. A language model, human analyst or hyperparameter search may use the measured *development* history repeatedly. Let T be an almost surely finite stopping time for that history, and condition on $\mathcal{D} = \mathcal{F}_T$. Let the now fixed finite family have $M(\mathcal{D}) = C(\mathcal{D})n(\mathcal{D})$ positive coordinates, with $\gamma(\mathcal{D}) = \alpha/M(\mathcal{D})$. Assume for every fixed pair that its *new final* count has conditional law $K_{\theta j} \mid \mathcal{D} \sim \mathrm{Binom}(B_{\theta j}, r_j(\theta))$, where $B_{\theta j}$, the release, tolerance, sampling frame and attack were fixed by $\mathcal{D}$ and the final draws were not used to construct it. By the exact inversion of Appendix D,

$$\Pr\{r_j(\theta) > \overline{r}_{\theta j} \mid \mathcal{D}\} \leq \gamma(\mathcal{D}), \qquad \Pr\{\exists(\theta, j) : r_j(\theta) > \overline{r}_{\theta j} \mid \mathcal{D}\} \leq \alpha. \tag{81}$$

The second inequality is a union bound *conditional* on the complete adaptive development history; the number and identity of pairs may depend on that history. On its complementary event, every accepted candidate satisfies every declared tolerance. Integrating over $\mathcal{D}$ proves an unconditional error probability no greater than α. No claim follows for scenarios omitted from the declared family or for a final audit that has influenced development. In particular, feeding a final audit result back to an LLM and then trying another candidate with the same nominal α is an unaccounted additional selection step.

Why measured trials cannot be replaced by language-model labels. The proper-scoring identity in (57) assumes $\mathbb{E}[K_{ij} \mid X_i = x, B_{ij} = b] = b\pi_j(x)$. Replacing K_{ij} by a model-generated estimate $\widetilde{K}_{ij}$ changes the minimiser to $\mathbb{E}[\widetilde{K}_{ij}/B_{ij} \mid X_i = x]$, which need not equal $\pi_j(x)$. A prompt that asserts a small risk is therefore no statistical evidence of small

risk, even when its prose is plausible. The software's proposal and outcome formats enforce a procedural distinction; source integrity and experimental independence still require institutional checks.

Sensitivity versus validity. For a fixed network with widths q_m, the spectral-norm bound

$$\|\widehat{\boldsymbol{r}}(x) - \widehat{\boldsymbol{r}}(x')\|_2 \leq \frac{1}{4}\left(\prod_{m=1}^{L+1} \|W_m\|_2\right) \|x - x'\|_2 \tag{82}$$

uses ReLU's unit Lipschitz constant and sigmoid's derivative at most 1/4. The bound is for *predictions* at fixed weights. It provides no bound on $\|\widehat{\boldsymbol{r}} - \boldsymbol{r}\|$ or on risk under new auxiliary information. The certification statement is instead obtained from fresh final audit data and the declared experimental law.

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